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By Amit Katiyar (MCA-JNU)



MATHEMATICS

- If $\left(\frac{15}{8}\right) + \left(\frac{15}{7}\right) = \left(\frac{n}{r}\right)$, then the values of n and r are
(a) 16 and 7 (b) 16 and 8
(c) 16 and 9 (d) 30 and 15
- In a class of 50 students, it was found that 30 students read "Hitavad", 35 students read "Hindustan" and 10 read neither. How many students read both "Hitavad" and "Hindustan" newspaper?
(a) 25 (b) 20 (c) 15 (d) 30
- If $A = \{4^x - 3x - 1 : x \in N\}$ and $B = \{9(x - 1) : x \in N\}$, where N is the set of natural numbers, then
(a) $A \subset B$ (b) $A \subseteq B$ (c) $A \supset B$ (d) $A \supseteq B$
- If $A = \{x, y, z\}$, then the number of subsets in powerset of A is
(a) 6 (b) 8 (c) 7 (d) 9
- How many word can be formed starting with letter D taking all letters from word DELHI so that the letters are not repeated?
(a) 4 (b) 12 (c) 24 (d) 120
- Naresh has 10 friends and he wants to invite 6 of them to a party. How many times will 3 particular friends never attend the party?
(a) 8 (b) 7 (c) 720 (d) 35
- There is a young boy's birthday party in which 3 friends have attended. The mother has arranged 10 games where a prize is awarded for a winning game. The prizes are indential. If each of the 4 children receives at least one prize, then how many distribution of prizes are possible?
(a) 80 (b) 84 (c) 70 (d) 72
- Three cities A, B, C are equidistant from each other. A motorist travels from A to B at 30 km/h, from B to C at 40 km/h and from C to A at 50 km/h. Then the average speed is
(a) 39 km/h (b) 40 km/h (c) 38.3 km/h (d) 37.6 km/h
- A problem in Mathematics is given to 3 students A, B and C. If the probability of A solving the problem is $\frac{1}{2}$ and B not solving it is $\frac{1}{4}$. The whole probability of the problem being solved is $\frac{63}{64}$, then what is the probability of solving it by C?
(a) $\frac{1}{8}$ (b) $\frac{1}{64}$ (c) $\frac{7}{8}$ (d) $\frac{1}{2}$
- A and B play a game where each is asked to select a number from 1 to 25. If the two numbers match, both win a prize. The probability that they will not win a prize in a single trial is
(a) $\frac{1}{25}$ (b) $\frac{24}{25}$ (c) $\frac{2}{25}$ (d) $\frac{3}{25}$
- A, B, C are three sets of values of x
A: 2, 3, 7, 1, 3, 2, 3
B: 7, 5, 9, 1, 2, 5, 3, 8
C: 4, 4, 1, 1, 7, 2, 3, 4
Select the correct statement among the following
(a) Mean of A is equal to mode of C.
(b) Mean of C is equal to Median of B.
(c) Median of B is equal to mode of A.
(d) Mean, Median and Mode of A are same.
- Standard deviation for the following distribution is
Size of item : 6 7 8 9 10 11 12
Frequency: 3 6 9 13 8 5 4
(a) 1.607 (b) 9.0 (c) 5.0 (d) 1.88

- If $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$, then for any positive integer n, A^n is
(a) $\begin{bmatrix} \sin n\alpha & \cos n\alpha \\ \cos n\alpha & -\sin n\alpha \end{bmatrix}$ (b) $\begin{bmatrix} \cos n\alpha & \sin n\alpha \\ \sin n\alpha & \cos n\alpha \end{bmatrix}$
(c) $\begin{bmatrix} \cos n\alpha & \sin n\alpha \\ \sin n\alpha & -\cos n\alpha \end{bmatrix}$ (d) $\begin{bmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{bmatrix}$
- Roots of equation $ax^2 - 2bx + c = 0$ are n and m, then the value of $\frac{b}{an^2+c} + \frac{b}{am^2+c}$ is
(a) $\frac{c}{a}$ (b) $\frac{b}{a}$ (c) $\frac{a}{c}$ (d) $\frac{b}{c}$
- The number of values of k for which the linear equations
 $4x + ky + z = 0$
 $kx + 4y + z = 0$
 $2x + 2y + z = 0$
possess a non-zero solution is
(a) 2 (b) 1 (c) 0 (d) 3
- Let $A = (a_{ij})$ and $B = (b_{ij})$ be two square matrices of order n and $\det(A)$ denotes the determinant of A. Then, which of the following is not correct?
(a) If A is a diagonal matrix, then $\det(A) = a_{11}a_{22}\dots a_{nn}$.
(b) $\det(AB) = \det(A)\det(B)$
(c) $\det(cA) = c \det(A)$
(d) $\det(A) = \det(A^T)$ where A^T denotes the transpose of the matrix A.
- The tangent to an ellipse $x^2 + 16y^2 = 16$ and making angle 60° with X-axis is:
(a) $x - \sqrt{3}y + 7 = 0$ (b) $\sqrt{3}x - y + 8 = 0$
(c) $\sqrt{3}x - y + 7 = 0$ (d) $x + \sqrt{3}y - 7 = 0$
- Find the number of point(s) of intersection of the ellipse $\frac{x^2}{4} + \frac{(y-1)^2}{9} = 1$ and the circle $x^2 + y^2 = 4$
(a) 4 (b) 3 (c) 2 (d) 1
- An arithmetic progression has 3 as its first term. Also, the sum of the first 8 terms is twice the sum of the first 5 terms. Then what is the common difference?
(a) $\frac{3}{4}$ (b) $\frac{1}{2}$ (c) $\frac{1}{4}$ (d) $\frac{4}{3}$
- If $a+b+c=0$, then the value of $\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab}$ is
(a) 1 (b) 0 (c) 3 (d) -1
- $\lim_{x \rightarrow 0} x^2 e^{\sin(\frac{1}{x})}$
(a) 1 (b) limit does not exist
(c) infinity (d) None of these
- If $f(x) = \begin{cases} x^2, & x \leq 0 \\ 2\sin x, & x > 0 \end{cases}$ then $x=0$ is
(a) Point of minima (b) Point of maxima
(c) Point of discontinuity (d) None of these
- If $g(x) = \begin{cases} \frac{x^2-x}{2x}, & x \neq 0 \\ k, & x = 0 \end{cases}$ is a continuous function at $x=0$, then the value of k is
(a) 2 (b) $\frac{1}{2}$ (c) 1 (d) None
- Find the interval(s) on which the graph $y = 2x^3 e^x$ is increasing
(a) $(-3, 0)$ and $(0, \infty)$ (b) $(-3/2, 0)$ and $(0, \infty)$
(c) $(-3, \infty)$ only (d) None of these
- If $I = \int \sec^2 x \operatorname{cosec}^4 x dx = -\frac{1}{3} \cot^3 x + k \tan x - 2 \cot x + C$, the value of k is
(a) 1 (b) 2 (c) 3 (d) 4
- Evaluate $\int e^x \left(\frac{1+\sin x \cos x}{\cos^2 x} \right) dx$
(a) $e^x \cos x + c$ (b) $e^x \sec x \tan x + c$
(c) $e^x \tan x + c$ (d) $e^x \cos^2 x - 1 + c$



27. If $I_n = \int_0^a (a^2 - x^2)^n dx$ where n is a positive integer, then the relation between I_n and I_{n-1} is

- (a) $I_n = \frac{2na^2}{2n+1} I_{n-1}$ (b) $I_n = \frac{2n^2 a^2}{2n+1} I_{n-1}$
 (c) $I_n = \frac{2na^2}{2n-1} I_{n-1}$ (d) $I_n = \frac{2n^2 a^2}{2n-1} I_{n-1}$

28. The value of $I = \int_{-2}^2 (ax^5 + bx^3 + c) dx$ depends on the

- (a) Value of b (b) Value of c
 (c) Value of a (d) Value of a and b

29. Find the area bounded by the line $y = 3 - x$, the parabola $y = x^2 - 9$ and $x \geq -1, y \geq 0$.

- (a) $7/2$ (b) $11/2$ (c) $9/2$ (d) None

30. If a, b, c are three non-coplanar vectors, then

- $(a+b+c) \cdot [(a+b) \times (a+c)] =$
 (a) 0 (b) $[a \ b \ c]$ (c) $2[a \ b \ c]$ (d) $-[a \ b \ c]$

31. Two forces F_1 and F_2 are used to pull a car, which met an accident. The angle between the two forces is θ . Find the value of θ for which the resultant force is equal to $\sqrt{F_1^2 + F_2^2}$

- (a) $\theta = 0^\circ$ (b) $\theta = 45^\circ$ (c) $\theta = 90^\circ$ (d) $\theta = 135^\circ$

32. If a, b, c, d are four vectors such that $a+b+c$ is Collinear with d and $b+c+d$ is collinear with a , then $a+b+c+d$ is

- (a) 0 (b) collinear with $a+d$
 (c) collinear with $a-d$ (d) collinear with $b-c$

33. Forces of magnitude 5, 3, 1 units act in the directions $6\hat{i} + 2\hat{j} + 3\hat{k}$, $3\hat{i} - 2\hat{j} + 6\hat{k}$, $2\hat{i} - 3\hat{j} - 6\hat{k}$ respectively on a particle which is displaced from the point $(2, -1, -3)$ to $(5, -1, 1)$. The total work done by the force is

- (a) 21 units (b) 5 units (c) 33 units (d) 105 units

34. The position vectors of points A and B are a and b . then the position vector of point P dividing AB in the ratio $m:n$ is

- (a) $\frac{na+mb}{m+n}$ (b) $\frac{na+mb}{m-n}$ (c) $\frac{na-mb}{m+n}$ (d) None

35. If a, b, c are three non-zero vectors with no two of which are collinear, $a+2b$ is collinear with c and $b+3c$ is collinear with a , then $|a+2b+6c|$ will be equal to

- (a) zero (b) 9 (c) 1 (d) None

36. Vertices of the vectors $\hat{i} - 2\hat{j} + 2\hat{k}$, $2\hat{i} + \hat{j} - \hat{k}$ and $3\hat{i} - \hat{j} + 2\hat{k}$ form a triangle, this triangle is

- (a) Equilateral triangle
 (b) Right angle triangle
 (c) Two side are equal in length
 (d) None of the above

37. If the volume of a parallelopiped whose adjacent edges $a = 2\hat{i} + 3\hat{j} + 4\hat{k}$, are $b = \hat{i} + a\hat{j} + 2\hat{k}$ is 15, then $\alpha = c = \hat{i} + 2\hat{j} + a\hat{k}$

- (a) 1 (b) $5/2$ (c) $9/2$ (d) 0

38. Solve the equation $\sin^2 x - \sin x - 2 = 0$ for x on the interval $0 \leq x < 2\pi$

- (a) $x = -\frac{\pi}{2}$ only (b) $x = \frac{\pi}{4}$ and $\frac{2\pi}{7}$
 (c) $x = \frac{2\pi}{3}$ and $\frac{2\pi}{5}$ (d) None of these

39. If $\frac{\tan x}{2} = \frac{\tan y}{3} = \frac{\tan z}{5}$ and $x + y + z = \pi$, then the value of $\tan^2 x + \tan^2 y + \tan^2 z$ is

- (a) $38/3$ (b) 38 (c) 114 (d) None

40. Find the value of $\sin 12^\circ \sin 48^\circ \sin 54^\circ$

- (a) $1/8$ (b) $1/6$ (c) $1/2$ (d) $1/4$

41. If $\cos x = \tan y$, $\cot y = \tan z$ and $\cot z = \tan x$, then $\sin x =$

- (a) $\frac{\sqrt{5}-1}{2}$ (b) $\frac{\sqrt{5}+1}{2}$ (c) $\frac{\sqrt{5}-1}{4}$ (d) $\frac{\sqrt{5}+1}{4}$

42. The value of $\tan\left(45 + \frac{\theta}{2}\right)$ is

- (a) $\tan \theta - \sec \theta$ (b) $\tan \theta + \sec \theta$
 (c) $\cot \theta - \sec \theta$ (d) $\cot \theta + \sec \theta$

43. The value of $\sin 10^\circ \sin 50^\circ \sin 70^\circ$ is

- (a) $1/4$ (b) $1/2$ (c) $3/4$ (d) $1/8$

44. The expression $\frac{\tan A}{1-\cot A} + \frac{\cot A}{1-\tan A}$ can be written as

- (a) $\sin A \cos A + 1$ (b) $\sec A \operatorname{cosec} A + 1$
 (c) $\tan A + \cot A$ (d) $\sec A + \operatorname{cosec} A$

45. Angle of elevation of the top of the tower from 3 points (collinear) A, B and C on a road leading to the foot of the tower are 30° , 45° and 60° , respectively. The ratio of AB and BC is

- (a) $\sqrt{3}:1$ (b) $\sqrt{3}:2$
 (c) $1:2$ (d) $2:\sqrt{3}$

46. The area enclosed between the curves $y^2 = x$ and $y = |x|$ is

- (a) $2/3$ sq. unit (b) 1 sq. unit
 (c) $1/6$ sq. unit (d) $1/3$ sq. unit

47. Test the continuity of the function at $x=2$

$$f(x) = \begin{cases} \frac{5}{2} - x, & x < 2 \\ x - \frac{3}{2}, & x > 2 \end{cases}$$

- (a) Continuous at $x=2$ (b) Discontinuous at $x=2$
 (c) Semicontinuous at $x=2$ (d) None of these

48. The value of $2 \tan^{-1}[\operatorname{cosec}(\tan^{-1} x) - \tan(\cot^{-1} x)]$ is

- (a) $\tan x$ (b) $\cot x$ (c) $\tan^{-1} x$ (d) $\operatorname{cosec}^{-1} x$

49. If $3 \sin x + 4 \cos x = 5$, then $6 \tan \frac{x}{2} - 9 \tan^2 \frac{x}{2} =$

- (a) 1 (b) 3 (c) 4 (d) 6

50. If A is a subset of B and B is a subset of C, then cardinality of $A \cup B \cup C$ is equal to

- (a) cardinality of C (b) cardinality of B
 (c) cardinality of A (d) None of these

ANALYTICAL ABILITY & LOGICAL REASONING

51. A set of consecutive positive integers beginning with 1 is written on the blackboard. A student came along and erased one number. The average of the remaining numbers is 35717. What was the number erased?

- (a) 7 (b) 8 (c) 9 (d) None

52. Four friends A, B, C and D need to cross a bridge in the night. A maximum of 2 people can cross at a time. they have only one lamp. A takes one min to cross the bridge. B Take 2 min, C takes 8 min and D takes 11 min to cross the bridge respectively. A pair must walk together at the speed of the person who walks slowly. What is the minimum time required to cross the bridge by all the four people?

- (a) 23 min (b) 20 min (c) 18 min (d) 16 min

53. In a city, 40.1% of the adults are illiterate while 85.1% the children are literate. If the ratio of the adults to that the children is 2:3, then what percent of the population is literate?

- (a) 20% (b) 25% (c) 50% (d) 75%

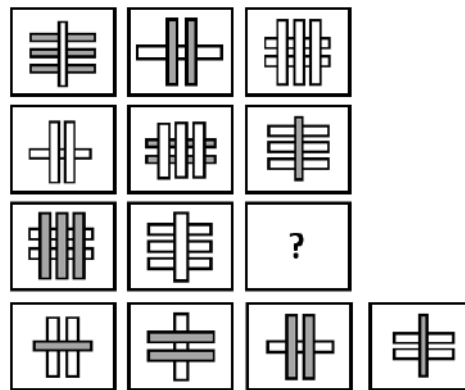
54. A runs $1\frac{2}{3}$ times as fast as B. If A gives B a start of 80m, how far must winning post be so that A and B might reach it at the same time?

- (a) 200m (b) 400m (c) 300m (d) 160m



55. A person's present age is two fifth of the age of his mother. After 8 year, he will be one half of the (age) of his mother. What is the present age of his mother?
(a) 60yr (b) 50yr (c) 40yr (d) 30yr
56. Mr. Kumar drives to work at an average speed of 48 km/hr. the time taken to cover the first 60% of the distance is 10 min more than the time taken to cover the remaining distance. How far is his office ?
(a) 30km (b) 40km (c) 45km (d) 48km
57. Two pipes A and B can fill the cistern in 37.5 min and 45 min respectively. Both pipes are opened. The cistern will be filled in just half an hour, if the B is turned off after
(a) 5 min (b) 9 min (c) 10 min (d) 15 min
58. In a certain code, DOES is written as 5\$3% and SITE is written as %4#3. How is EDIT written in that code?
(a) 3#4\$ (b) %3#5 (c) 354# (d) 4#5\$
59. In a shower, 5cm of rain falls. The volume of water that falls on 1.5 hectares of ground is
(a) 75m³ (b) 750m³ (c) 7500m³ (d) 75000m³
60. Eight friends A through H, are sitting around a circular table, playing a game of cards. They belong to two different teams X and Y. No two persons of the same team sit in adjacent seats.
- A sits neither opposite to D nor to H but is sitting in between C and G.
 - B sits neither opposite to A nor to G but is sitting in between F and D.
 - B and H belong to team X and D sits opposite to E
- Who are the members of team X?
(a) A, D, F and E (b) B, H, C and E
(c) B, D, H and G (d) B, H, C and G
61. Eight friends A through H, are sitting around a circular table, playing a game of cards. They belong to two different teams X and Y. No two persons of the same team sit in adjacent seats.
- A sits neither opposite to D nor to H but is sitting in between C and G.
 - B sits neither opposite to A nor to G but is sitting in between F and D
 - B and H belong to team X and D sits opposite to E
- Who are sitting adjacent to E?
(a) B and H (b) B and G (c) H and G (d) H and C
62. Four students A, B, C and D distributed 30 marbles among themselves. No two students got equal number of marbles. No students got more than 10 marbles. No student got less than 5 marbles. A and C got odd number of marbles. B and D got even number of marbles. A got more marbles than B, C got more marbles than D, B got more marbles than D.
What is the number of marbles with A?
(a) 6 (b) 7 (c) 8 (d) 9
63. Four students A, B, C and D distributed 30 marbles among themselves. No two students got equal number of marbles. No students got more than 10 marbles. No student got less than 5 marbles. A and C got odd number of marbles. B and D got even number of marbles. A got more marbles than B, C got more marbles than D, B got more marbles than D.
Mean of number of marbles with B, C, D is
(a) 6 (b) 7 (c) 8 (d) None

64. Nine individuals-Z, Y, X, W, V, U, T, S and R- are the only candidates, who can serve on three committees-k₁, k₂ and k₃, and each candidate should serve on exactly one of the committees, Committee k₁ should consist of exactly one number more than committee-k₂. It is possible that there are no members in committee k₃. Among Z, Y and X none can serve on committee k₁. Among W, V and U none can serve on committee k₂. Among T, S and R none can serve on committee k₃.
In case committee K₂ is served by T and Z only, how many of the nine individuals should serve on committee k₃?
(a) 3 (b) 4 (c) 5 (d) 6
65. Nine individuals – Z, Y, X, W, V, U, T, S and R- are the only candidates, who can serve on three committees- K₁, K₂ and K₃, and each candidate should serve on exactly one of the committees. Committee K₁ should consist of exactly one member more than committee K₂. It is possible that there are no members in committee K₃. Among Z, Y and X none can serve on committee K₁, Among W, V and U none can serve on committee K₂.
Among T, S and R none can serve on committee K₃.
Of the nine individuals, the largest number that can serve together on committee K₃ is
(a) 8 (b) 7 (c) 6 (d) 5
66. Fill in the blank in the series ELFA, GLHA, ILJA, _____, MLNA:
(a) OLPA (b) KLMA (c) LLMA (d) KLLA
67. Pointing to a gentleman, Mohan said, 'His only brother is the father of my daughter's father'. The gentleman is Mohan's
(a) brother (b) father (c) uncle (d) None
68. It was 9.35 am in Garvita 's watch, which kept correct time, when Manya informed her that the last bus left the bus stop at 9.25 am. Manya' s watch is 5 min fast. The frequency of the bus is every 20 min. For how long Garvita must wait to catch the next bus?
(a) 5 min (b) 10 min (c) 15 min (d) 20 min
69. A total of 324 notes comprising of ₹20 and ₹ 50 denomination make a sum of ₹ 12450. The number of ₹ 20 notes is
(a) 200 (b) 144 (c) 125 (d) 110
70. Rishabh stops after going 10 km towards west from his office. Then he goes 8 km turning to his left. After this he goes 4 km turning to his left. How far is he from the fixed point?
(a) 18 km (b) 8 km (c) 10 km (d) None
71. Which of the four options should fill the missing cell?



- (a) (b) (c) (d)
(a)1 (b)2 (c)3 (d)4





72. In the following questions, the symbols \$, #, @, % and * illustrate the following meanings.

- > P\$Q – P is not smaller than Q
- > P#Q – P is neither greater than nor equal to Q.
- > P@Q – P is neither smaller than nor equal to Q.
- > P%Q – P is not greater than Q
- > P*Q – P is neither greater than nor smaller than Q.

Statements: K#L, L%M, M*N, N#O

Conclusions:

- I. K#M II. K*M III. L%O
- (a) I only (b) Either I or II only
- (c) III only (d) All I, II and III

73. Study the following arrangement carefully and answer the question given below:

W1R % 4JE # 7MT2I9BH3A\$9FQ5DG6USP

Three of the following are alike in a certain way on the basis of above arrangement and hence form a group.

Which one does not belong to that group?

- (a) R W 4 (b) 9 Q A (c) 3 B \$ (d) 5 F G

74. If there are no dancers that aren't slim no singers that aren't dancers, then which statements are always true?

Choose the correct answer.

- (a) There is not one slim person that is not a dancer.
- (b) All singers are slim.
- (c) Anybody slim is also a singer.
- (d) None of the above

75. If in a certain language, ITNIETAM is the code for INTIMATE, which word has the code TREVNIETARBI?

- (a) INVERTIBRATE (b) INVERTIBARTE
- (c) INVERTIBRATE (d) INVERTIBRETA

76. Sum of ages of Anu and bhanu is 10 yr more than sum of ages of Bhanu, Chanu and Dhanu. Average age of Chanu and dhanu is 19 yrs . Find the average age of Anu and Dhanu if Dhanu is 10 years elder than Chanu.

- (a) 36 yr (b) 30 yr
- (c) 25 yr (d) 31 yr

77. In a competitive examination in Maharashtra state 9% candidates got selected from the total appeared candidates. Tripura state had an equal number candidates appeared and 12% candidates got selected with 102 more candidates got selected than Maharashtra state. What was the number of candidates appeared from each state?

- (a) 3400 (b) 3000
- (c) 2850 (d) 3200

78. Shiva gave 40% of his monthly salary to his mother from the remaining he used 7% for electronic gadgets and 23% he kept aside for his monthly expenses. The remaining amount he transferred to his friend's account. The sum of the amount he gave to his mother and he transferred to his friend account was ₹ 41000. What was Shiva's monthly salary?

- (a) ₹ 50500 (b) ₹ 49000
- (c) ₹ 50000 (d) ₹ 45000

79. Read the information given below and answer the questions that follow:

- (i) A*B means → A and B are of the same age
 - (ii) A-B means → B is younger than A
 - (iii) A+ B means → A is younger than B
- Sachin * Madan- Reena means?
- (a) Reena is youngest
 - (b) Reena is oldest
 - (c) Madan is younger than Reena
 - (d) Madan is youngest

80. Read the information given below and answer the questions that follows

- (i) A*B means → A and B are of the same age
 - (ii) A-B means → B is younger than A
 - (iii) A+ B means → A is younger than B
- X + Y + Z is same as.....?

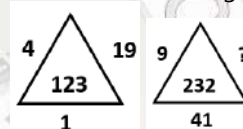
- (a) Y-X-Z (b) Z-Y-X (c) Z-X-Y (d) X-Y-Z

81. Find out the wrong number in the following number series:

56, 58, 62, 70, 84, 118, 182

- (a) 58 (b) 62 (c) 84 (d) 118

82. Find out the missing number



- (a) 2 (b) 6 (c) 4 (d) 16

83. In a examination, 78% of the total students who appeared were successful. If the total number of failures was 176 and 34% got first class, then how many students got first class?

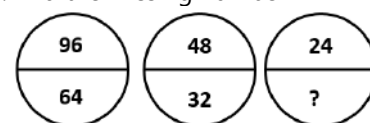
- (a) 272 (b) 112 (c) 210 (d) 254

84. Which number should come in place of the question mark(?) in the following chart:

1	7	9
2	14	?
3	105	117

- (a) 16 (b) 26 (c) 20 (d) 12

85. Find the missing number



- (a) 21 (b) 16 (c) 10 (d) 8

86. How many minimum numbers of colours will be required to paint all the sides of a cube without the adjacent sides having the same colours?

- (a) 3 (b) 4 (c) 5 (d) 6

87. If a man walks at the rate of 4km/hr, he misses a train by only 6 min. However, if he walks at the rate of 5km/hr, he reaches the station 6 min before the arrival of the train. The distance covered by him reach the station is

- (a) 4 km (b) 7 km (c) 9 km (d) 5 km

88. If the numerator of a fraction is increased by 25% and denominator decreased by 20%, the new value is 5/4. What is the original value?

- (a) 3/5 (b) 4/5 (c) 7/8 (d) 3/7



89. Read the following information carefully and answer the questions given below

- Five friends Amar, Kapil, Sarvesh, Rohan and Nagesh put on five shirts of different colour, i.e., Red, Yellow, Blue, White, and Green, while they were going to attend to party. These colours are not in the order.
- They have different hobbies as reading, playing, outing singing and writing.
- Kapil, who like singing, does not wear Yellow shirt. Sarvesh wear Red shirt and he does not like reading or writing. Nagesh likes playing and he does not wear Blue or Yellow shirt. Amar likes writing and Rohan does not wear Yellow or Green shirt.

What is the colour of Kapil's shirt?

- White
- Green
- Blue
- Insufficient data to answer

90. Read the following information carefully and answer the questions given below

- Five friends Amar, Kapil, Sarvesh, Rohan and Nagesh put on five shirts of different colour, i.e., Red, Yellow, Blue, White, and Green, while they were going to attend a party. These colours are not in a order.
- They have different hobbies as reading, playing, outing, singing and writing.
- Kapil, who likes singing, does not wear Yellow shirt, Sarvesh wear Red shirt and he does not like reading or writing. Nagesh likes playing and he does not wear Blue or Yellow shirt. Amar likes writing and Rohan does not wear Yellow or green shirt.

Who likes writing?

- Rohan
- Amar
- Kapil
- Insufficient data to answer

GENERAL ENGLISH

91. Assume 'x' represents negation of x the Boolean function $x'y' + xy + x'y$ is equivalent to?

- $x' + y$
- $x + y$
- $x + y'$
- $x' + y'$

92. The memory unit which directly communicates with the CPU is known as

- Primary Memory
- Secondary Memory
- Shared Memory
- Auxiliary Memory

93. Dynamic RAM consumespower andthan Static RAM

- More, Faster
- More, slower
- Less, Slower
- Less, Faster

94. The binary equivalent of $(234.125)_{10}$

- $(11101010.101)_2$
- $(10101010.011)_2$
- $(11101010.001)_2$
- $(10101110.011)_2$

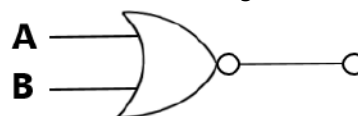
95. Determine the octal equivalent of $(432267)_{10}$?

- $(432267)_8$
- $(346731)_8$
- $(2164432)_8$
- None of these

96. One Exabyte is equal to...

- 1018 Bytes
- 1 Zetta Bytes divided (/) by one thousand
- 1 Peta Bytes multiplied (x) by one thousand
- All of the above

97. Consider the following circuit.



How many minimum numbers of two input NAND gates are required to design the above circuit ?

- 6
- 4
- 5
- 3

98. The time required for fetching and execution of one simple machine instruction is known as

- Delay time
- CPU cycle
- Real time
- Seek Time

99. The equivalence of given expression $x + x'y$ with Boolean theorem is.....

- x
- $x + y$
- x'
- 0

100. The logic XOR operation of $(4A(0))_{16}$ and $(B53F)_{16}$ results

- AACB
- 0000
- FFFF
- ABCD

GENERAL ENGLISH

101. Choose the correct expression of approval

- Super!
- Rotten!
- Damn!
- Hell!

102. Which of the following is a noun?

- Carelessness
- Careless
- Carelessly
- Caring

103. Choose the word that accurately signifies a student who avoids attending classes.

- Diligent
- Callous
- Morose
- Truant

104. Identify the type of error in the following sentence Some of the books, were destroyed.

- Syntactical error
- Punctuation error
- Grammatical error
- Conflicting error

105. Pick the word similar in meaning ALLEVIATE

- Clear
- Lessen
- Match
- Incite

106. Pick the word opposite in meaning ABSURD

- Cruel
- Sensible
- Calm
- Sturdy

107. Identify the meaning of the following

It was all greek to me.....

- Difficult to speak
- Difficult to write
- Difficult to arrange
- Difficult to understand

108. "To vouch for" means

- To confirm
- To degrade
- To follow
- To supersede

109. "To hold your horses" means

- To be ready
- To be patient
- To be eager
- To be impatient

110. Choose the right option.

Blessing in disguise is?

- Something good
- Something unrecognised
- Something known to all
- Something good but not recognised at first



MAARULA CLASSES

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NIMCET-2020

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111. He was accused theft.

- (a) on (b) about
- (c) in (d) of

112. I never listen..... the radio.

- (a) to (b) of
- (c) about (d) in

113. I don't think I' ve ever.... On that sofa.

- (a) been sitting (b) sat
- (c) sit (d) sitting

114. Name the letter that is sent along with the CV (Curriculum Vitae).

- (a) Formal letter (b) Covering letter
- (c) Introductory (d) Business letter

115. What is not included in a resume?

- (a) Work experience (b) Education
- (c) Projects (d) Family history

116. Choose the correct sentence of the following

- (a) I prefer coffee to tea.
- (b) I prefer coffee for tea.
- (c) I prefer coffee than tea.
- (d) I prefer coffee by tea.

Read the following passage and answer the questions

A Lichen is a composite organism that arises from algae living among filaments of multiple fungi in a symbiotic relationship. The combined Lichen has properties different from those of its component organisms. Lichens come in many colours, sizes, and forms. The properties are sometimes plant like, but Lichens are not plants. Lichens may have tiny leafless branches, flat leaflike structures or flackes that lie on the surface like peeling paint or other growth forms.

Lichens occur from sea level to high alpine elevations, in many environmental conditions and can grow on almost any surface. Different kinds of Lichens have adopted to survive in some of the most extreme environment on earth such as Arctic, Tundra, hot dry deserts, rocky coasts, and toxic slagheaps. They can even live inside solid rocks, growing between the grains.

It is estimated that 6% of the earth's land surface is - covered by Lichens. Some of them are considered to be the oldest living things. They are among the first living things to grow on fresh rock exposed after an event such as a land slide. The long-life span and slow but regular growth rate of some Lichens can be used to date events.

117. The passage states all the following about Lichens EXCEPT

- (a) Lichen is an independent plant.
- (b) Lichens have different properties.
- (c) Lichens can grow in exotic conditions.
- (d) Lichens can be used to date events.

118. The passage aims at the view.

- (a) that Lichens are toxic in nature.
- (b) that sharing of things help easy growth.
- (c) that Lichens should be excluded from Botany
- (d) how plants use solar energy.

119. Choose the one which best expresses the meaning of the word

FLACKES

- (a) Peeling (b) Pip
- (c) Loaf (d) Whole

120. Identify the one word opposite to SPAN in meaning

- (a) Stretch (b) Length
- (c) Duration (d) Compress





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MAARULA MATHEM –E LEARNING

ANSWER KEY

1. (b)	21. (d)	41. (a)	61. (d)	81. (c)	101. (a)
2. (a)	22. (d)	42. (b)	62. (d)	82. (c)	102. (a)
3. (a)	23. (d)	43. (d)	63. (b)	83. (a)	103. (d)
4. (b)	24. (c)	44. (b)	64. (b)	84. (d)	104. (b)
5. (c)	25. (a)	45. (a)	65. (c)	85. (b)	105. (b)
6. (b)	26. (c)	46. (c)	66. (d)	86. (a)	106. (b)
7. (b)	27. (a)	47. (b)	67. (c)	87. (a)	107. (d)
8. (c)	28. (b)	48. (c)	68. (a)	88. (b)	108. (a)
9. (c)	29. (d)	49. (a)	69. (c)	89. (d)	109. (b)
10. (b)	30. (d)	50. (a)	70. (c)	90. (b)	110. (d)
11. (d)	31. (c)	51. (a)	71. (a)	91. (a)	111. (d)
12. (a)	32. (a)	52. (c)	72. (a)	92. (a)	112. (a)
13. (d)	33. (c)	53. (d)	73. (b)	93. (c)	113. (b)
14. (d)	34. (a)	54. (a)	74. (d)	94. (c)	114. (b)
15. (c)	35. (a)	55. (c)	75. (c)	95. (d)	115. (d)
16. (c)	36. (b)	56. (b)	76. (a)	96. (d)	116. (a)
17. (c)	37. (c)	57. (b)	77. (a)	97. (b)	117. (a)
18. (b)	38. (d)	58. (c)	78. (c)	98. (b)	118. (b)
19. (a)	39. (a)	59. (b)	79. (a)	99. (b)	119. (a)
20. (c)	40. (a)	60. (d)	80. (b)	100. (c)	120. (d)

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website: www.question.maarula.in



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MATHS SOLUTION

Calculations:

$$\binom{15}{8} + \binom{15}{7} = \binom{n}{r}$$

$$\Rightarrow {}^{15}C_8 + {}^{15}C_7 = {}^nC_r$$

$$\text{Using } {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r:$$

$$\Rightarrow {}^{15}C_8 + {}^{15}C_{8-1} = {}^{15+1}C_8 = {}^nC_r$$

$$\Rightarrow {}^{16}C_8 = {}^nC_r$$

$$\Rightarrow n = 16 \text{ and } r = 8$$

2. Let A be the set of students who read "Hitavard" and B the set of students who read "Hindustan".

From the given information:

$$n(A \cup B) = 50 - 10 = 40.$$

$$n(A) = 30.$$

$$n(B) = 35.$$

$$\text{using } n(A \cup B) = n(A) + n(B) - n(A \cap B), \text{ we get:}$$

$$40 = 30 + 35 - n(A \cap B)$$

$$\Rightarrow n(A \cap B) = 25.$$

$\therefore$ The number of students who read both "Hitavard" and "Hindustan" newspapers is $n(A \cap B) = 25$.

3. We know that a set of which all the elements are contained in another set is known as a **subset**

Calculations:

Given, $X = \{4^n - 3n - 1 : n \in \mathbb{N}\}$, where $\mathbb{N}$ is the set of natural numbers so $N = 1, 2, 3, \dots$

After putting all these values of N , we will get the elements of X .

$$\Rightarrow X = \{0, 9, 54, \dots\}$$

Similarly, $Y = \{9(n - 1) : n \in \mathbb{N}\}$, where $\mathbb{N}$ is the set of natural numbers,

$$\Rightarrow Y = \{0, 9, 18, 27, \dots\}$$

Here, all elements of X is contained in set Y .

Hence, X is a subset of Y .

$$X \subset Y \subset X$$

Hence, if $x = \{4n - 3n - 1 : n \in \mathbb{N}\}$ and $Y = \{9(n - 1) : n \in \mathbb{N}\}$, where $\mathbb{N}$ is the set of natural numbers, then $X \subset Y \subset X$.

- 4.. **Calculations:**

Given, $A = \{x, y, z\}$.

The power set (or powerset) of a Set A is defined as the set of all subsets of the Set A including the Set itself and the null or empty set.

Powerset of $A = \{\phi, \{x\}, \{y\}, \{z\}, \{x, y\}, \{y, z\}, \{x, z\}, \{x, y, z\}\}$.

Hence, the number of subsets in the powerset of A is 8.

5. **Calculations:**

The number of letters in the word DELHI = 5.

Since, D has to be the first letter, the number of ways in which the first letter can be written is 1.

The remaining 4 letters can be written in any order/arrangement of the remaining 4 letters (ELHI).

$$\therefore \text{The number of ways of writing the 4 letters} = 4! = 4 \times 3 \times 2 \times 1 = 24.$$

$$\therefore \text{The total number of ways of writing the 5 letters} = 1 \times 24 = 24.$$

6. ${}^nC_r = \frac{n!}{r!(n-r)!}$

Calculations:

Given, Naresh has 10 friends, and he wants to invite 6 of them to a party.

Remove the 3 particular friends and invite 6 friends from the remaining $= 10 - 3 = 7$ friends.

This can be done in 7C_6 ways

$$\text{Hence, required number of ways} = {}^7C_6$$

$$\Rightarrow \text{required number of ways} = \frac{7!}{(7-6)!6!}$$

$$\Rightarrow \text{required number of ways} = 7$$

Hence, Naresh has 10 friends, and he wants to invite 6 of them to a party. total number of ways that 3 particular friends never attend the party = 7

7. Here, the number of objects is $n = 10$ and the number of people is $r = 4$. Since, every person must get at least one object, the total number of possible distributions is:

$$\begin{aligned} {}^{n-1}C_{r-1} &= {}^{10-1}C_{4-1} = {}^9C_3 \\ &= \frac{9!}{3!(9-3)!} = \frac{7 \times 8 \times 9}{3 \times 2 \times 1} \\ &= 84 \end{aligned}$$

8. **Calculations:**

Let's say that the distance between each pair of cities is 600 km.

From A to B:

Distance = 600 km.

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}} = \frac{600}{30} = 20 \text{ hrs.}$$

From B to C:

Distance = 600 km.

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}} = \frac{600}{40} = 15 \text{ hrs.}$$

From C to A:

Distance = 600 km.

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}} = \frac{600}{50} = 12 \text{ hrs.}$$

For overall journey:

$$\text{Total distance} = 600 + 600 + 600 = 1800 \text{ km.}$$

$$\text{Total time} = 20 + 15 + 12 = 47 \text{ hrs.}$$

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}} = \frac{1800}{47} = 38.29 \text{ km/hr.}$$

9. It is given that $P(A) = \frac{1}{2}$ and $P(\text{not } B) = \frac{1}{4}$. Let's say that the probability of C solving the problem is $P(C) = x$.

$$\therefore \text{Probability of A not solving the problem} = P(\text{not } A) = 1 - P(A) = 1 - \frac{1}{2} = \frac{1}{2}.$$

$$\text{And, probability of C not solving the problem} = P(\text{not } C) = 1 - P(C) = 1 - x.$$

Now, the probability that the problem is not solved at all:

$$= P(\text{not } A) \text{ AND } P(\text{not } B) \text{ AND } P(\text{not } C)$$

$$= P(\text{not } A) \times P(\text{not } B) \times P(\text{not } C)$$

$$= \frac{1}{2} \times \frac{1}{4} \times (1 - x)$$

$$= \frac{1-x}{8}$$

And, the probability that the problem is solved $= 1 - \text{Probability that the problem is not solved at all} = \frac{63}{64}$.

And, the probability that the problem is solved $= 1 - \text{Probability that the problem is not solved at all} = \frac{63}{64}$.

$$\Rightarrow 1 - \frac{1-x}{8} = \frac{63}{64}$$

$$\Rightarrow \frac{1-x}{8} = 1 - \frac{63}{64} = \frac{1}{64}$$

$$\Rightarrow 1 - x = \frac{1}{8}$$

$$\Rightarrow x = 1 - \frac{1}{8} = \frac{7}{8}$$

$$\therefore \text{The probability of C solving the problem is } P(C) = x = \frac{7}{8}.$$

10. **Calculations:**

Given, A and B play a game where each is asked to select a number from 1 to 25. If the two numbers match, both win a prize.

Total number of possibilities = 25×25

Favourable number of their winning = 25

$$\Rightarrow \text{Probability of winning the prize} = \frac{1}{25}$$

The probability that they will not win a prize in a single trial = $1 - \text{The probability that they will win a prize in a single trial}$

The probability that they will not win a prize in a single trial = $1 - \frac{1}{25}$

$$\Rightarrow \text{The probability that they will not win a prize in a single trial} = \frac{24}{25}$$

11. There are 7 observations in each set, so the Median will be the value of observation at the $\left(\frac{7+1}{2}\right) = 8$ th position, when they are arranged in ascending/descending order.

The values of Mean, Median and Mode for the given sets is summarized below:





Ascending Order	Sum	Mean	Median	Mode
A : 1, 2, 2, 3, 3, 3, 7	21	3	3	3
B : 3, 5, 5, 7, 8, 9, 12	49	7	7	5
C : 2, 3, 4, 4, 4, 7, 11	35	5	4	4

'Mean, Median and Mode of A are same.' is correct.

12. Total number of items in the distribution

$$= \sum f_i = 3 + 6 + 9 + 13 + 8 + 5 + 4 = 48.$$

The Mean ($\bar{x}$) of the given set = $\frac{\sum f_i x_i}{\sum f_i}$

$$\Rightarrow \bar{x} = \frac{6 \times 3 + 7 \times 6 + 8 \times 9 + 9 \times 13 + 10 \times 8 + 11 \times 5 + 12 \times 4}{48} = \frac{432}{48} = 9$$

Let's calculate the variance using the formula: $\sigma^2 = \frac{\sum x_i^2}{n} - \bar{x}^2$.

$$\frac{\sum x_i^2}{n} = \frac{6^2 \times 3 + 7^2 \times 6 + 8^2 \times 9 + 9^2 \times 13 + 10^2 \times 8 + 11^2 \times 5 + 12^2 \times 4}{48} = \frac{4012}{48} = 83.58$$

$$\therefore \sigma^2 = 83.58 - 9^2 = 83.58 - 81 = 2.58$$

$$\text{And, Standard Deviation } (\sigma) = \sqrt{\sigma^2} = \sqrt{\text{Variance}} = \sqrt{2.58} \approx 1.607.$$

$$\text{Let } M = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \text{ and } N = \begin{bmatrix} \cos \beta & \sin \beta \\ -\sin \beta & \cos \beta \end{bmatrix}$$

$$\Rightarrow M \times N = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \times \begin{bmatrix} \cos \beta & \sin \beta \\ -\sin \beta & \cos \beta \end{bmatrix}$$

$$\Rightarrow M \times N = \begin{bmatrix} \cos \alpha \cos \beta - \sin \alpha \sin \beta & \cos \alpha \sin \beta + \sin \alpha \cos \beta \\ -\sin \alpha \cos \beta - \cos \alpha \sin \beta & -\sin \alpha \sin \beta + \cos \alpha \cos \beta \end{bmatrix}$$

$$\Rightarrow M \times N = \begin{bmatrix} \cos(\alpha + \beta) & \sin(\alpha + \beta) \\ -\sin(\alpha + \beta) & \cos(\alpha + \beta) \end{bmatrix}$$

$$\therefore A \times A = A^2 = \begin{bmatrix} \cos(\alpha + \alpha) & \sin(\alpha + \alpha) \\ -\sin(\alpha + \alpha) & \cos(\alpha + \alpha) \end{bmatrix} = \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ -\sin 2\alpha & \cos 2\alpha \end{bmatrix}$$

$$\Rightarrow A^2 \times A = A^3 = \begin{bmatrix} \cos(2\alpha + \alpha) & \sin(2\alpha + \alpha) \\ -\sin(2\alpha + \alpha) & \cos(2\alpha + \alpha) \end{bmatrix} = \begin{bmatrix} \cos 3\alpha & \sin 3\alpha \\ -\sin 3\alpha & \cos 3\alpha \end{bmatrix}$$

And so on

$$\text{Hence, } A^n = \begin{bmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{bmatrix}.$$

14. Given: $ax^2 - 2bx + c = 0$

Using the formula for the sum and the product of the roots we get:

$$m + n = \frac{2b}{a}$$

$$mn = \frac{c}{a}$$

Now, the given expression $\frac{b}{an^2+c} + \frac{b}{am^2+c}$:

$$= \frac{\frac{2b}{a}}{2\left(\frac{n^2}{a} + \frac{c}{a}\right)} + \frac{\frac{2b}{a}}{2\left(\frac{m^2}{a} + \frac{c}{a}\right)}$$

$$= \frac{m+n}{2(n^2+mn)} + \frac{m+n}{2(m^2+mn)}$$

$$= \frac{2n(n+m)}{2n(n+m)} + \frac{2m(m+n)}{2m(m+n)}$$

$$= \frac{1}{2} \left(\frac{1}{m} + \frac{1}{n} \right)$$

$$= \frac{1}{2} \left(\frac{m+n}{mn} \right)$$

$$= \frac{1}{2} \left(\frac{2b}{\frac{c}{a}} \right)$$

$$= \frac{b}{c}.$$

15. For the given set of equations:

$$4x + ky + z = 0$$

$$kx + 4y + z = 0$$

$$2x + 2y + z = 0$$

It can be observed that $x = y = z = 0$ is one solution of the system.

In order to have infinitely many (including non-zero) solutions, D must be zero.

$$\Rightarrow D = \begin{vmatrix} 4 & k & 2 \\ k & 4 & 2 \\ 1 & 1 & 1 \end{vmatrix} = 0$$

$$\Rightarrow 4(4 - 2) + k(2 - k) + 2(k - 4) = 0$$

$$\Rightarrow 8 + 2k - k^2 + 2k - 8 = 0$$

$$\Rightarrow k^2 - 4k = 0$$

$$\Rightarrow k(k - 4) = 0$$

$$\Rightarrow k = 0 \text{ OR } k - 4 = 0$$

$$\Rightarrow k = 0 \text{ OR } k = 4$$

Hence, there are 2 possible values of k but a non-zero solution is one.

16. From the properties of determinants stated above, it can be seen

that **$\det(cA) = c [\det(A)]$ is not correct.**

The correct statement would be $\det(cA) = c^n [\det(A)]$

17. The given tangent makes an angle of 60° with the positive direction of the x -axis.

$$\therefore m = \tan 60^\circ = \sqrt{3}$$

Converting the given equation of the ellipse into the standard

$$\text{form } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1:$$

$$x^2 + 16y^2 = 16$$

$$\Rightarrow \frac{x^2}{4^2} + \frac{y^2}{2^2} = 1$$

$$\therefore a = 4 \text{ and } b = 2$$

Using the formula $y = mx \pm \sqrt{a^2 m^2 + b^2}$ for the tangents to the ellipse, the required equation is:

$$y = \sqrt{3}x + \sqrt{4^2(\sqrt{3})^2 + 2^2}$$

$$\Rightarrow y = \sqrt{3}x + 7$$

$$\Rightarrow \sqrt{3}x - y + 7 = 0$$

18. Let $f(x, y) = \frac{x^2}{4} + \frac{(y-1)^2}{9} - 1 = 0$ and $g(x, y) = x^2 + y^2 - 4 = 0$ intersect at a point (a, b) .

$$\therefore f(a, b) = g(a, b) = 0$$

$$\Rightarrow \frac{a^2}{4} + \frac{(b-1)^2}{9} - 1 = a^2 + b^2 - 4 = 0$$

$$\Rightarrow a^2 + \frac{4(b-1)^2}{9} - 4 = a^2 + b^2 - 4 = 0$$

$$\Rightarrow \frac{4(b-1)^2}{9} = b^2$$

$$\Rightarrow 4(b-1)^2 = 9b^2$$

$$\Rightarrow (2b-2)^2 - (3b)^2 = 0$$

$$\Rightarrow (2b-2+3b)(2b-2-3b) = 0$$

$$\Rightarrow (5b-2)(-b-2) = 0$$

$$\Rightarrow 5b-2 = 0 \text{ OR } -b-2 = 0$$

$$\Rightarrow b = \frac{2}{5} \text{ OR } b = -2$$

Now, using $a^2 + b^2 - 4 = 0$, we get:

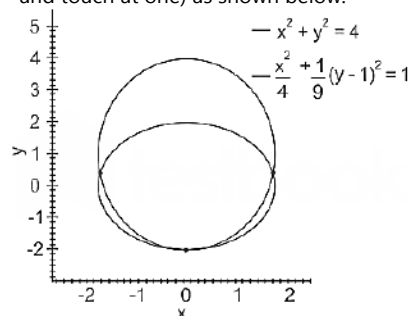
$$a^2 + \left(\frac{2}{5}\right)^2 - 4 = 0 \text{ OR } a^2 + (-2)^2 - 4 = 0$$

$$\Rightarrow a^2 = \frac{96}{25} \text{ OR } a^2 = 0$$

$$\Rightarrow a = \pm \frac{4\sqrt{6}}{5} \text{ OR } a = 0$$

$\therefore$ The points of intersection are $\left(\frac{4\sqrt{6}}{5}, \frac{2}{5}\right)$, $\left(-\frac{4\sqrt{6}}{5}, \frac{2}{5}\right)$ and $(0, -2)$.

The given ellipse and the circle **intersect at three points** (intersect at two and touch at one) as shown below:



19. It is given that first term $a = 3$. Let the common difference be d .

Also, the sum of the first 8 terms is twice the sum of the first 5 terms.



$$\Rightarrow S_8 = 2S_5$$

$$\Rightarrow \frac{8}{2}[3 + \{3 + (8-1)d\}] = 2 \times \frac{5}{2}[3 + \{3 + (5-1)d\}]$$

$$\Rightarrow 4(6 + 7d) = 5(6 + 4d)$$

$$\Rightarrow 24 + 28d = 30 + 20d$$

$$\Rightarrow 8d = 6$$

$$\Rightarrow d = \frac{3}{4}$$

∴ The common difference of the given arithmetic progression is $\frac{3}{4}$.

20. Since, $a + b + c = 0$, we have $a^3 + b^3 + c^3 = 3abc$.

∴ The given expression is:

$$\begin{aligned} & \frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab} \\ &= \frac{a^3}{abc} + \frac{b^3}{bca} + \frac{c^3}{cab} \\ &= \frac{3abc}{abc} = 3 \end{aligned}$$

21. We know that $-1 \leq \sin \theta \leq 1$.

$$\Rightarrow -1 \leq \sin\left(\frac{1}{x}\right) \leq 1$$

Since, e^x is a strictly increasing function for all real values of x , we can say that:

$$\Rightarrow e^{-1} \leq e^{\sin\left(\frac{1}{x}\right)} \leq e^1$$

Also, since $x^2 \geq 0$, we can say that:

$$\Rightarrow x^2 e^{-1} \leq x^2 e^{\sin\left(\frac{1}{x}\right)} \leq x^2 e^1$$

$$\Rightarrow \frac{x^2}{e} \leq x^2 e^{\sin\left(\frac{1}{x}\right)} \leq x^2 e$$

So, we can consider $h(x) = \frac{x^2}{e}$, $f(x) = \lim_{x \rightarrow 0} x^2 e^{\sin\left(\frac{1}{x}\right)}$ and $g(x) = x^2 e$.

$$\text{Now, } \lim_{x \rightarrow 0} h(x) = \lim_{x \rightarrow 0} \frac{x^2}{e} = 0.$$

$$\text{And } \lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} x^2 e = 0$$

Since, $\lim_{x \rightarrow 0} h(x) = \lim_{x \rightarrow 0} g(x) = 0$, we must have $\lim_{x \rightarrow 0} f(x) = 0$.

$$\text{Hence, } \lim_{x \rightarrow 0} x^2 e^{\sin\left(\frac{1}{x}\right)} = 0.$$

22. Let us check for the continuity and differentiability (maxima/minima) of the function at x

Continuity:

$$f(x) = \begin{cases} x^2; & x \leq 0 \\ 2\sin x; & x > 0 \end{cases}$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} x^2 = 0^2 = 0$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} 2\sin x = 2\sin 0 = 0$$

$$f(0) = 0^2 = 0$$

∴ $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0)$, the function $f(x)$ is continuous at $x = 0$.

Differentiability:

$$f'(x) = \begin{cases} 2x; & x \leq 0 \\ 2\cos x; & x > 0 \end{cases}$$

$$\lim_{x \rightarrow 0^-} f'(x) = \lim_{x \rightarrow 0^-} 2x = 2 \times 0 = 0$$

$$\lim_{x \rightarrow 0^+} f'(x) = \lim_{x \rightarrow 0^+} 2\cos x = 2\cos 0 = 2$$

∴ $\lim_{x \rightarrow 0^-} f'(x) \neq \lim_{x \rightarrow 0^+} f'(x)$, the function is not differentiable at $x = 0$.

23. For $x \neq 0$, the given function can be re-written as:

$$f(x) = \begin{cases} \frac{1-x}{2}; & x \neq 0 \\ K; & x = 0 \end{cases}$$

Since the equation of the function is same for $x < 0$ and $x > 0$, we have:

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{1-x}{2} = \frac{1-0}{2} = \frac{1}{2}$$

For the function to be continuous at $x = 0$, we must have:

$$\lim_{x \rightarrow 0} f(x) = f(0)$$

$$\Rightarrow K = \frac{1}{2}$$

24. Let us first find $\frac{dy}{dx}$ for the given function $y = 2x^3 e^x$.

$$\frac{d}{dx}(2x^3 e^x) = 2x^3 \left(\frac{d}{dx} e^x\right) + 2e^x \left(\frac{d}{dx} x^3\right)$$

$$= 2x^3 e^x + 6x^2 e^x$$

$$= 2x^2 e^x (x + 3)$$

At the extrema, $\frac{dy}{dx} = 0$.

$$\Rightarrow 2x^2 e^x (x + 3) = 0$$

Since, $e^x > 0$ for all values of x , we get:

$$\Rightarrow x^2 = 0 \text{ OR } x + 3 = 0$$

$$\Rightarrow x = 0 \text{ OR } x = -3$$

So, we need to analyze the function in the intervals $(-\infty, -3)$, $(-3, 0)$ and $(0, \infty)$.

In the interval $(0, \infty)$:

$$f(1) = 2x^2 e^x (x + 3) = 8e > 0$$

∴ $f(x)$ is increasing in $(0, \infty)$.

In the interval $(-3, 0)$:

$$f(-1) = 2(-1)^2 e^{-1} (-1 + 3) = \frac{4}{e} > 0$$

∴ $f(x)$ is increasing in $(-3, 0)$.

In the interval $(-\infty, -3)$:

$$f(-4) = 2(-4)^2 e^4 (-4 + 3) = -\frac{32}{e^4} < 0.$$

∴ $f(x)$ is decreasing in $(-\infty, -3)$.

We can see that the function is increasing in the intervals $(-3, 0)$ and $(0, \infty)$.

25. Integrating by parts by taking $\csc^4 x$ as the first function and $\sec^2 x$ as the second function:

$$\int \sec^2 x \csc^4 x dx$$

$$= \csc^4 x \int \sec^2 x dx - \int \left[\left(\frac{d}{dx} \csc^4 x \right) \left(\int \sec^2 x dx \right) \right] dx + C$$

$$= \csc^4 x \tan x - \int 4 \csc^3 x (-\cot x \csc x) \tan x dx + C$$

$$= \csc^4 x \tan x + 4 \int \csc^4 x dx + C$$

Integrating $\csc^4 x$ by parts and taking $\csc^2 x$ as the first and second functions:

$$\int (\csc^2 x)(\csc^2 x) dx$$

$$= \csc^2 x \int \csc^2 x dx - \int \left[\left(\frac{d}{dx} \csc^2 x \right) \left(\int \csc^2 x dx \right) \right] dx + C$$

$$= \csc^2 x (-\cot x) - \int 2 \csc x (-\cot x \csc x) (-\cot x) dx + C$$

$$= -\csc^2 x \cot x - 2 \int \csc^2 x \cot^2 x dx + C$$

Integrating $\csc^2 x \cot^2 x$ by parts and taking $\cot^2 x$ as the first and $\csc^2 x$ as the second for

$$\int \csc^2 x \cot^2 x dx$$

$$= \cot^2 x \int \csc^2 x dx - \int \left[\left(\frac{d}{dx} \cot^2 x \right) \left(\int \csc^2 x dx \right) \right] dx + C$$

$$= \cot^2 x (-\cot x) - \int 2 \cot x (-\csc^2 x) (-\cot x) dx + C$$

$$= -\cot^3 x - 2 \int \csc^2 x \cot^2 x dx + C$$

$$\Rightarrow 3 \int \csc^2 x \cot^2 x dx = -\cot^3 x + C$$

$$\Rightarrow \int \csc^2 x \cot^2 x dx = -\frac{\cot^3 x}{3} + C$$

Finally, $\int \sec^2 x \csc^4 x dx$

$$= \csc^4 x \tan x + 4 \left[-\csc^2 x \cot x - 2 \left(-\frac{\cot^3 x}{3} \right) \right] + C$$

$$= \csc^4 x \tan x - 4 \csc^2 x \cot x + \frac{8}{3} \cot^3 x + C$$

$$= (1 + \cot^2 x)^2 \tan x - 4(1 + \cot^2 x) \cot x + \frac{8}{3} \cot^3 x + C$$

$$= (1 + 2\cot^2 x + \cot^4 x) \tan x - 4\cot x - 4\cot^3 x + \frac{8}{3} \cot^3 x + C$$

$$= \tan x + 2\cot x + \cot^3 x - 4\cot x - \frac{4}{3} \cot^3 x + C$$

$$= -\frac{1}{3} \cot^3 x + \tan x - 2\cot x + C$$

∴ Comparing $\left(-\frac{1}{3} \cot^3 x + \tan x - 2\cot x + C \right)$

with $\left(-\frac{1}{3} \cot^3 x + k \tan x - 2\cot x + C \right)$ we can say that $k = 1$.

$$\int e^x \left(\frac{1 + \sin x \cos x}{\cos^2 x} \right) dx$$

$$= \int e^x \left(\frac{1}{\cos^2 x} + \frac{\sin x \cos x}{\cos^2 x} \right) dx$$

$$26. = \int e^x [\sec^2 x + \tan x] dx$$

$$= \int e^x [f(x) + f(x)] dx, \text{ where } f(x) = \tan x$$

$$= e^x f(x) + C$$

$$= e^x \tan x + C$$

27. Let us first evaluate $\int (a^2 - x^2)^n dx$ by integrating it in parts. Let's say

$$f(x) = (a^2 - x^2)^n \text{ and } g(x) = 1.$$

$$\therefore I = \int [(a^2 - x^2)^n \times 1] dx$$

$$\Rightarrow I = (a^2 - x^2)^n \int 1 dx - \int [n(a^2 - x^2)^{n-1} \times 2x \int 1 dx] dx$$

$$\Rightarrow I = x(a^2 - x^2)^n - 2n \int x^2 (a^2 - x^2)^{n-1} dx$$

$$\Rightarrow I = x(a^2 - x^2)^n + 2n \int (a^2 - x^2 - a^2)(a^2 - x^2)^{n-1} dx$$

$$\Rightarrow I = x(a^2 - x^2)^n + 2n \left[\int (a^2 - x^2)^n dx - a^2 \int (a^2 - x^2)^{n-1} dx \right]$$

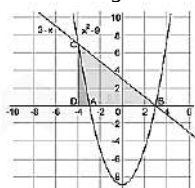
Since $I = \int (a^2 - x^2)^n dx$, we get:



$$\begin{aligned} \Rightarrow I &= x(a^2 - x^2)^n + 2nl - 2na^2 \int (a^2 - x^2)^{n-1} dx \\ \Rightarrow (1 - 2n)I &= x(a^2 - x^2)^n - 2na^2 \int (a^2 - x^2)^{n-1} dx \\ \Rightarrow I &= \frac{x(a^2 - x^2)^n}{1 - 2n} + \frac{2na^2}{2n-1} \int (a^2 - x^2)^{n-1} dx \\ \Rightarrow I_n &= \left[\frac{x(a^2 - x^2)^n}{1 - 2n} \right]_0^a + \frac{2na^2}{2n-1} \int_0^a (a^2 - x^2)^{n-1} dx \\ \Rightarrow I_n &= \left(\frac{2na^2}{2n-1} \right) I_{n-1} \end{aligned}$$

28. We observe that ax^5 and bx^3 are odd functions of x ,
 $\therefore a(-x)^5 = -ax^5$ and $b(-x)^3 = -bx^3$
 $\therefore \int_{-2}^2 ax^5 dx = 0$ and $\int_{-2}^2 bx^3 dx = 0$
 The value of $\int_{-2}^2 (ax^5 + bx^3 + c) dx$ depends on the value of c .
 In fact,
 $\int_{-2}^2 (ax^5 + bx^3 + c) dx = \int_{-2}^2 c dx$
 $= c[2 - (-2)]$
 $= 4c$

29. Let's say that the two curves are $f(x, y) = x + y - 3 = 0$
 and $g(x, y) = x^2 - y - 9 = 0$.
 The points of their intersection are the points where $f(x, y) = g(x, y)$.
 $\Rightarrow x + y - 3 = x^2 - y - 9 = 0$
 $\Rightarrow -x - y + 3 = x^2 - y - 9 = 0$
 $\Rightarrow x^2 + x - 12 = 0$
 $\Rightarrow x^2 + 4x - 3x - 12 = 0$
 $\Rightarrow x(x + 4) - 3(x + 4) = 0$
 $\Rightarrow (x + 4)(x - 3) = 0$
 $\Rightarrow x + 4 = 0$ OR $x - 3 = 0$
 $\Rightarrow x = -4$ OR $x = 3$.
 And, $y = 3 - (-4) = 7$ OR $y = 3 - 3 = 0$.
 Hence, the curves intersect at the points B(3, 0) and C(-4, 7) as shown in the diagram below:



The points where $y = x^2 - 9$ cuts the x -axis
 ($y = 0$) are A(-3, 0) and B(3, 0).
 The required area is the shaded part ABC = Area of BDC - Area of ADC
 $= \int_{-4}^3 (3 - x) dx - \int_{-4}^3 (x^2 - 9) dx$
 $= 3[x]_{-4}^3 - \left[\frac{x^2}{2} \right]_{-4}^3 - \left[\frac{x^3}{3} \right]_{-4}^3 + 9[x]_{-4}^3$
 $= 3[3 - (-4)] - \frac{1}{2}[3^2 - (-4)^2] - \frac{1}{3}[(-3)^3 - (-4)^3] + 9[-3 - (-4)]$
 $= 3(7) - \frac{1}{2}(-7) - \frac{1}{3}(37) + 9(1)$
 $= 21 + \frac{7}{2} - \frac{37}{3} + 9$
 $= \frac{127}{6}$

The answer is None of these.

30. $S = (\vec{a} + \vec{b} + \vec{c}) \cdot [(\vec{a} + \vec{b}) \times (\vec{a} + \vec{c})]$
 $\Rightarrow S = (\vec{a} + \vec{b} + \vec{c}) \cdot [(\vec{a} \times \vec{a} + \vec{a} \times \vec{c} + \vec{b} \times \vec{a} + \vec{b} \times \vec{c})]$
 $\Rightarrow S = (\vec{a} + \vec{b} + \vec{c}) \cdot [(0 + \vec{a} \times \vec{c} + \vec{b} \times \vec{a} + \vec{b} \times \vec{c})]$
 $S = \vec{a} \cdot [\vec{a} \times \vec{c} + \vec{b} \times \vec{a} + \vec{b} \times \vec{c}] + \vec{b} \cdot [\vec{a} \times \vec{c} + \vec{b} \times \vec{a} + \vec{b} \times \vec{c}] + \vec{c} \cdot [\vec{a} \times \vec{c} + \vec{b} \times \vec{a} + \vec{b} \times \vec{c}]$
 $S = \vec{a} \cdot (\vec{a} \times \vec{c}) + \vec{a} \cdot (\vec{b} \times \vec{a}) + \vec{a} \cdot (\vec{b} \times \vec{c}) + \vec{b} \cdot (\vec{a} \times \vec{c}) + \vec{b} \cdot (\vec{b} \times \vec{a}) + \vec{b} \cdot (\vec{b} \times \vec{c})$
 $+ \vec{c} \cdot (\vec{a} \times \vec{c}) + \vec{c} \cdot (\vec{b} \times \vec{a}) + \vec{c} \cdot (\vec{b} \times \vec{c})$
 $\Rightarrow S = 0 + 0 + \vec{a} \cdot (\vec{b} \times \vec{c}) + \vec{b} \cdot (\vec{a} \times \vec{c}) + 0 + 0 + 0 + \vec{c} \cdot (\vec{b} \times \vec{a}) + 0$
 $\Rightarrow S = \vec{a} \cdot (\vec{b} \times \vec{c}) + \vec{b} \cdot (\vec{a} \times \vec{c}) + \vec{c} \cdot (\vec{b} \times \vec{a})$
 $\Rightarrow S = \vec{a} \cdot (\vec{b} \times \vec{c}) + [-\vec{b} \cdot (\vec{c} \times \vec{a})] + [-\vec{c} \cdot (\vec{a} \times \vec{b})]$
 $\Rightarrow S = [\vec{a}, \vec{b}, \vec{c}] - [\vec{a}, \vec{b}, \vec{c}] - [\vec{a}, \vec{b}, \vec{c}]$
 $\therefore S = -[\vec{a}, \vec{b}, \vec{c}]$

31. Given:
 Resultant of forces F_1 and $F_2 = |\vec{F}_1 + \vec{F}_2|$
 $\Rightarrow \sqrt{F_1^2 + F_2^2} = \sqrt{F_1^2 + F_2^2 + 2 F_1 F_2 \cos \theta}$
 $\Rightarrow F_1^2 + F_2^2 = F_1^2 + F_2^2 + 2 F_1 F_2 \cos \theta$
 $\Rightarrow 0 = 2 F_1 F_2 \cos \theta$
 $\Rightarrow \cos \theta = 0$
 $\Rightarrow \theta = 90^\circ$

32. (a) Given:

$$\begin{aligned} \vec{a} + \vec{b} + \vec{c} &\text{ is collinear with } \vec{d} \\ \Rightarrow (\vec{a} + \vec{b} + \vec{c}) \times \vec{d} &= 0 \\ \Rightarrow (\vec{a} + \vec{b} + \vec{c}) \times \vec{d} + \vec{d} \times \vec{d} &= 0 (\because \vec{d} \times \vec{d} = 0) \\ \Rightarrow (\vec{a} + \vec{b} + \vec{c} + \vec{d}) \times \vec{d} &= 0 \end{aligned}$$

Also given:

$$\begin{aligned} \vec{b} + \vec{c} + \vec{d} &\text{ is collinear with } \vec{a} \\ \Rightarrow (\vec{b} + \vec{c} + \vec{d}) \times \vec{a} &= 0 \\ \Rightarrow \vec{a} \times \vec{a} + (\vec{b} + \vec{c} + \vec{d}) \times \vec{a} &= 0 (\because \vec{a} \times \vec{a} = 0) \\ \Rightarrow (\vec{a} + \vec{b} + \vec{c} + \vec{d}) \times \vec{a} &= 0 \end{aligned}$$

On adding (i) and (ii):

$$\begin{aligned} \Rightarrow (\vec{a} + \vec{b} + \vec{c} + \vec{d}) \times \vec{a} + (\vec{a} + \vec{b} + \vec{c} + \vec{d}) \times \vec{d} &= 0 \\ \Rightarrow (\vec{a} + \vec{b} + \vec{c} + \vec{d}) \times (\vec{a} + \vec{d}) &= 0 \end{aligned}$$

On subtracting (i) from (ii):

$$\begin{aligned} \Rightarrow (\vec{a} + \vec{b} + \vec{c} + \vec{d}) \times \vec{a} - (\vec{a} + \vec{b} + \vec{c} + \vec{d}) \times \vec{d} &= 0 \\ \Rightarrow (\vec{a} + \vec{b} + \vec{c} + \vec{d}) \times (\vec{a} - \vec{d}) &= 0 \end{aligned}$$

From (iii) and (iv), it is clear that $(\vec{a} + \vec{b} + \vec{c} + \vec{d})$ is collinear to $(\vec{a} + \vec{d})$ and $(\vec{a} - \vec{d})$

33. (c) The work done $W = (\vec{F}_r) \cdot (\vec{dx})$

where $\vec{F}_r$ is the resultant force, and $\vec{dx}$ is the displacement

The unit vector in the direction of $\vec{P} = \hat{P} = \frac{\vec{P}}{|\vec{P}|}$

Calculation:

The initial position of the particle (A) = $2\vec{i} - \vec{j} - 3\vec{k}$

The initial position of the particle (B) = $5\vec{i} - \vec{j} + \vec{k}$

Displacement $\vec{dx} = B - A = (5\vec{i} - \vec{j} + \vec{k}) - (2\vec{i} - \vec{j} - 3\vec{k}) = 3\vec{i} + 4\vec{k}$

First force F_1 = magnitude of the 1st force $\times$ unit vector in the given direction

$$\Rightarrow \vec{F}_1 = 5 \times \frac{6\vec{i} + 2\vec{j} + 3\vec{k}}{|6\vec{i} + 2\vec{j} + 3\vec{k}|}$$

$$\Rightarrow \vec{F}_1 = \frac{30\vec{i} + 10\vec{j} + 15\vec{k}}{7}$$

Second force F_2 = magnitude of the 2nd force $\times$ unit vector in the given direction

$$\Rightarrow \vec{F}_2 = 3 \times \frac{3\vec{i} - 2\vec{j} + 6\vec{k}}{|3\vec{i} - 2\vec{j} + 6\vec{k}|}$$

$$\Rightarrow \vec{F}_2 = \frac{2\vec{i} - 3\vec{j} - 6\vec{k}}{7}$$

Resultant Force $\vec{F}_r = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$

$$\Rightarrow \vec{F}_r = \frac{30\vec{i} + 10\vec{j} + 15\vec{k}}{7} + \frac{9\vec{i} - 6\vec{j} + 18\vec{k}}{7} + \frac{2\vec{i} - 3\vec{j} - 6\vec{k}}{7}$$

$$\Rightarrow \vec{F}_r = \frac{41\vec{i} + \vec{j} + 27\vec{k}}{7}$$

The work done by the forces:

$$W = (\vec{F}_r) \cdot (\vec{dx})$$

$$\Rightarrow W = \left(\frac{41\vec{i} + \vec{j} + 27\vec{k}}{7} \right) \cdot (3\vec{i} + 4\vec{k})$$

$$\Rightarrow W = \left(\frac{123 + 0 + 108}{7} \right) = \frac{231}{7}$$

$$\Rightarrow W = 33 \text{ units}$$

Calculation:

34. Assuming,

$$\vec{a} = x_a \vec{i} + y_a \vec{j} + z_a \vec{k}$$

$$\vec{b} = x_b \vec{i} + y_b \vec{j} + z_b \vec{k}$$

Vector of point p is

$$\vec{p} = x_p \vec{i} + y_p \vec{j} + z_p \vec{k}$$

$$\therefore x_p = \frac{nx_1 + mx_2}{n+m}, y_p = \frac{ny_1 + my_2}{n+m} \text{ and } z_p = \frac{nz_1 + mz_2}{n+m}$$

$$\Rightarrow \vec{p} = \frac{nx_a + mx_b}{n+m} \vec{i} + \frac{ny_a + my_b}{n+m} \vec{j} + \frac{nz_a + mz_b}{n+m} \vec{k}$$

$$\Rightarrow \vec{p} = \frac{n(x_a \vec{i} + y_a \vec{j} + z_a \vec{k}) + m(x_b \vec{i} + y_b \vec{j} + z_b \vec{k})}{n+m}$$

$$\Rightarrow \vec{p} = \frac{n\vec{a} + m\vec{b}}{n+m}$$

If a vector A is collinear to other vector B , it can be written as: $\vec{A} = \lambda \times \vec{B}$ where λ is any scalar number

35. Calculation:

Given $\vec{a} + 2\vec{b}$ is collinear with $\vec{c}$

$$\Rightarrow \vec{a} + 2\vec{b} = \lambda_1 \times \vec{c}$$

Also given $\vec{b} + 3\vec{c}$ is collinear with $\vec{a}$

$$\Rightarrow \vec{b} + 3\vec{c} = \lambda_2 \times \vec{a}$$

Subtracting $2 \times$ (ii) from (i)

$$\Rightarrow \vec{a} - 6\vec{c} = \lambda_1 \vec{c} - 2\lambda_2 \vec{a}$$

$$\therefore \lambda_1 = -6, \lambda_2 = \frac{-1}{2}$$



From (ii)

$$\Rightarrow \vec{b} + 3\vec{c} = \frac{-1}{2}\vec{a}$$

$$\Rightarrow 2\vec{b} + 6\vec{c} = -\vec{a}$$

$$\Rightarrow \vec{a} + 2\vec{b} + 6\vec{c} = 0$$

$$\Rightarrow |\vec{a} + 2\vec{b} + 6\vec{c}| = 0$$

36. Let's say that the vertices of the triangle are:

$$\vec{A} = i - 2j + 2k$$

$$\vec{B} = 2i + j - k$$

$$\vec{C} = 3i - j + 2k$$

$$\therefore \vec{AB} = \vec{B} - \vec{A} = (2i + j - k) - (i - 2j + 2k) = i + 3j - 3k$$

$$|\vec{AB}| = \sqrt{1^2 + 3^2 + (-3)^2} = \sqrt{19}$$

$$\vec{BC} = \vec{C} - \vec{B} = (3i - j + 2k) - (2i + j - k) = i - 2j + 3k$$

$$|\vec{BC}| = \sqrt{1^2 + (-2)^2 + 3^2} = \sqrt{14}$$

$$\vec{CA} = \vec{A} - \vec{C} = (i - 2j + 2k) - (3i - j + 2k) = -2i - j + 0k$$

$$|\vec{CA}| = \sqrt{(-2)^2 + (-1)^2 + 0^2} = \sqrt{5}$$

Since, $AB^2 = 19 = BC^2 + CA^2 = 14 + 5$, the triangle is right angled at C.

37. The sides of the parallelepiped are:

$$\vec{a} = 2i + 3j + 4k$$

$$\vec{b} = i + \alpha j + 2k$$

$$\vec{c} = i + 2j + \alpha k$$

$$\therefore \text{Volume} = [\vec{a} \vec{b} \vec{c}] = \begin{vmatrix} 2 & 3 & 4 \\ 1 & \alpha & 2 \\ 1 & 2 & \alpha \end{vmatrix} = 15$$

$$\Rightarrow 2(a^2 - 4) + 3(2 - \alpha) + 4(2 - \alpha) = 15$$

$$\Rightarrow 2a^2 - 8 + 6 - 3\alpha + 8 - 4\alpha = 15$$

$$\Rightarrow 2a^2 - 7\alpha - 9 = 0$$

$$\Rightarrow 2a^2 - 9a + 2a - 9 = 0$$

$$\Rightarrow a(2a - 9) + (2a - 9) = 0$$

$$\Rightarrow (2a - 9)(a + 1) = 0$$

$$\Rightarrow 2a - 9 = 0 \text{ OR } a + 1 = 0$$

$$\Rightarrow a = \frac{9}{2} \text{ OR } a = -1$$

38. $\sin^2 x - \sin x - 2 = 0$

Using the formula for the roots of a quadratic equation:

$$\Rightarrow \sin x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-2)}}{2(1)}$$

$$\Rightarrow \sin x = \frac{1 + \sqrt{9}}{2} \text{ OR } \sin x = \frac{1 - \sqrt{9}}{2}$$

$$\Rightarrow \sin x = 2 \text{ OR } \sin x = -1$$

$$\therefore -1 \leq \sin x \leq 1, \therefore \sin x = 2 \text{ is not possible.}$$

$$\sin x = -1 = -\sin \frac{\pi}{2} = \sin \left(-\frac{\pi}{2} \right) = \sin \left(2n\pi - \frac{\pi}{2} \right), n \in \mathbb{Z}$$

$$\text{For } n = 0, x = -\frac{\pi}{2}$$

$$\text{For } n = 1, x = \frac{3\pi}{2}$$

$$\text{For } n = 2, x = \frac{7\pi}{2}$$

The only value of x on the interval $0 \leq x < 2\pi$ is $x = \frac{3\pi}{2}$.

39. It is given that $\frac{\tan x}{2} = \frac{\tan y}{3} = \frac{\tan z}{5} = k$ (say).

$$\therefore \tan x = 2k, \tan y = 3k \text{ and } \tan z = 5k.$$

It is also given that $x + y + z = \pi$.

$$\Rightarrow x + y = \pi - z$$

$$\Rightarrow \tan(x + y) = \tan[\pi - z]$$

$$\Rightarrow \frac{\tan x + \tan y}{1 - \tan x \tan y} = \tan(-z) = -\tan z$$

$$\Rightarrow \tan x + \tan y = -\tan z + \tan x \tan y \tan z$$

$$\Rightarrow \tan x + \tan y + \tan z = \tan x \tan y \tan z$$

Substituting the values in terms of k from the above result, we get:

$$2k + 3k + 5k = (2k)(3k)(5k)$$

$$\Rightarrow 10k = 30k^3$$

$$\Rightarrow k^2 = \frac{1}{3}$$

$$\text{Now, } \tan^2 x + \tan^2 y + \tan^2 z$$

$$= (2k)^2 + (3k)^2 + (5k)^2$$

$$= 4k^2 + 9k^2 + 25k^2$$

$$= 38k^2$$

$$38 \times \frac{1}{3} = \frac{38}{3}$$

40.

$$\begin{aligned} & \sin 12^\circ \sin 48^\circ \sin 54^\circ \\ &= \frac{2 \sin 48^\circ \sin 12^\circ}{2} \sin 54^\circ \\ &= \left[\frac{\cos(48^\circ - 12^\circ) - \cos(48^\circ + 12^\circ)}{2} \right] \sin(90^\circ - 36^\circ) \\ &= \left(\frac{\cos 36^\circ - \cos 60^\circ}{2} \right) \cos 36^\circ \\ &= \left(\frac{1}{2} \right) \left(\frac{\sqrt{5}+1}{4} - \frac{1}{2} \right) \left(\frac{\sqrt{5}+1}{4} \right) \\ &= \left(\frac{1}{2} \right) \left(\frac{\sqrt{5}-1}{4} \right) \left(\frac{\sqrt{5}+1}{4} \right) \\ &= \left(\frac{1}{2} \right) \left(\frac{5-1}{16} \right) \\ &= \frac{1}{8} \end{aligned}$$

41. Calculation:

It is given that $\cos x = \tan y$, $\cot y = \tan z$ and $\cot z = \tan x$.

$$\therefore \cos x = \tan y = \frac{1}{\cot y} = \frac{1}{\tan z} = \cot z = \tan x = \frac{\sin x}{\cos x}$$

$$\Rightarrow \cos^2 x = \sin x$$

$$\Rightarrow 1 - \sin^2 x = \sin x$$

$$\Rightarrow \sin^2 x + \sin x - 1 = 0$$

Using the formula for the roots of a quadratic equation:

$$\Rightarrow \sin x = \frac{-1 \pm \sqrt{1^2 - 4(1)(-1)}}{2(1)}$$

$$\Rightarrow \sin x = \frac{-1 + \sqrt{5}}{2} \text{ OR } \sin x = \frac{-1 - \sqrt{5}}{2}$$

From the given answer options, $\sin x = \frac{\sqrt{5}-1}{2}$.

42.

$$\text{Let } \tan \left(45^\circ + \frac{\theta}{2} \right) = x$$

$$\Rightarrow x = \frac{\tan 45^\circ + \tan \frac{\theta}{2}}{1 - \tan 45^\circ \tan \frac{\theta}{2}}$$

$$\Rightarrow x = \frac{1 + \tan \frac{\theta}{2}}{1 - \tan \frac{\theta}{2}} \quad (\because \tan 45^\circ = 1)$$

$$\Rightarrow x = \frac{\cos \frac{\theta}{2} + \sin \frac{\theta}{2}}{\cos \frac{\theta}{2} - \sin \frac{\theta}{2}}$$

$$\Rightarrow x = \frac{\cos \frac{\theta}{2} + \sin \frac{\theta}{2}}{\cos \frac{\theta}{2} - \sin \frac{\theta}{2}} \times \frac{\cos \frac{\theta}{2} + \sin \frac{\theta}{2}}{\cos \frac{\theta}{2} + \sin \frac{\theta}{2}}$$

$$\Rightarrow x = \frac{\left(\cos \frac{\theta}{2} + \sin \frac{\theta}{2} \right)^2}{\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}}$$

$$\Rightarrow x = \frac{\cos^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2} + 2 \cos \frac{\theta}{2} \sin \frac{\theta}{2}}{\cos \theta}$$

$$\Rightarrow x = \frac{1 + \sin \theta}{\cos \theta}$$

$$\Rightarrow x = \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta} = \sec \theta + \tan \theta$$

43.

$$\begin{aligned} & \sin 10^\circ \sin 50^\circ \sin 70^\circ \\ &= \sin 10^\circ \sin(90^\circ - 40^\circ) \sin(90^\circ - 20^\circ) \\ &= \sin 10^\circ \cos 40^\circ \cos 20^\circ \\ &= \left(\frac{2 \sin 10^\circ \cos 10^\circ}{2 \cos 10^\circ} \right) \cos 40^\circ \cos 20^\circ \\ &= \left(\frac{\sin 20^\circ}{2 \cos 10^\circ} \right) \cos 40^\circ \cos 20^\circ \\ &= \left(\frac{2 \sin 20^\circ \cos 20^\circ}{4 \cos 10^\circ} \right) \cos 40^\circ \\ &= \frac{2 \sin 40^\circ \cos 40^\circ}{8 \cos 10^\circ} \\ &= \frac{\sin 80^\circ}{8 \cos 10^\circ} \\ &= \frac{\sin(90^\circ - 10^\circ)}{8 \cos 10^\circ} \\ &= \frac{\cos 10^\circ}{8 \cos 10^\circ} \\ &= \frac{1}{8} \end{aligned}$$



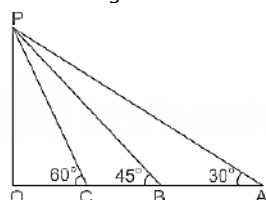
44.

Let $\tan A = m \Rightarrow \cot A = \frac{1}{m}$

$$\begin{aligned} \therefore \frac{\tan A}{1 - \cot A} + \frac{\cot A}{1 - \tan A} \\ = \frac{m}{1 - \frac{1}{m}} + \frac{\frac{1}{m}}{1 - m} \\ = \frac{m^2}{m-1} - \frac{1}{m(m-1)} \\ = \frac{m^3 - 1}{m(m-1)} \\ = \frac{(m-1)(m^2 + m + 1)}{m(m-1)} \\ = \frac{m^2 + m + 1}{m} \\ = m + 1 + \frac{1}{m} \\ = \left(\frac{\sin A}{\cos A} + \frac{\cos A}{\sin A} \right) + 1 \\ = \left(\frac{\sin^2 A + \cos^2 A}{\sin A \cos A} \right) + 1 \\ = \left(\frac{1}{\sin A \cos A} \right) + 1 \\ = \sec A \csc A + 1 \end{aligned}$$

45.

Let the length of the tower OP be x



From the figure,

$$OA = \frac{x}{\tan 30} = \sqrt{3}x$$

$$OB = \frac{x}{\tan 45} = x$$

$$OC = \frac{x}{\tan 60} = \frac{x}{\sqrt{3}}$$

$$\therefore AB = OA - OB = \sqrt{3}x - x$$

$$BC = OB - OC = x - \frac{x}{\sqrt{3}}$$

$$\Rightarrow \frac{AB}{BC} = \frac{\sqrt{3}x - x}{x - \frac{x}{\sqrt{3}}}$$

$$\Rightarrow \frac{AB}{BC} = \frac{\sqrt{3}(\sqrt{3}-1)}{\sqrt{3}-1}$$

$$\Rightarrow AB : BC = \sqrt{3} : 1$$

46. Let's first find the points of intersection/touching of the two

curves $y^2 = x$ and $y = |x|$.

Using the condition for intersecting, we must have:

$$y^2 - x = y - |x| = 0$$

CASE 1: For, $x \geq 0$, $|x| = x$.

$$\Rightarrow y^2 - x = y - x = 0$$

$$\Rightarrow y^2 - y = 0$$

$$\Rightarrow y(y - 1) = 0$$

$$\Rightarrow y = 0 \text{ OR } y - 1 = 0$$

$$\Rightarrow y = 0 \text{ OR } y = 1.$$

$$\text{And, } x = 0^2 = 0 \text{ OR } x = 1^2 = 1.$$

The points of intersection are (0, 0) and (1, 1).

CASE 2: For $x < 0$, $|x| = -x$.

$$\Rightarrow y^2 - x = y - (-x) = 0$$

$$\Rightarrow y^2 - x = -y - x = 0$$

$$\Rightarrow y^2 + y = 0$$

$$\Rightarrow y(y + 1) = 0$$

$$\Rightarrow y = 0 \text{ OR } y + 1 = 0$$

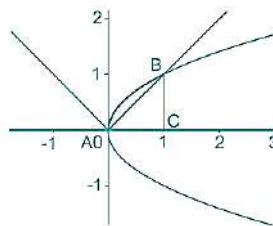
$$\Rightarrow y = 0 \text{ OR } y = -1.$$

$\therefore y = |x|$, y cannot be negative.

$\therefore$ The only solution in this case $y = 0$ and $x = 0^2 = 0$.

Finally, the solutions (points of intersection/touching) of $y^2 = x$ and $y = |x|$ are (0, 0) and (1, 1).

The graph of the curves is shown below:



The required (shaded) area is:

(Area under $y^2 = x$ from $x = 0$ to $x = 1$) - (Area under $y = |x|$ from $x = 0$ to $x = 1$).

$$= \left| \int_0^1 \sqrt{x} dx \right| - \left| \int_0^1 x dx \right|$$

$$= \left| \frac{2}{3} [x^{\frac{3}{2}}]_0^1 \right| - \left| \frac{1}{2} [x^2]_0^1 \right|$$

$$= \left| \frac{2}{3} [1^{\frac{3}{2}} - 0^{\frac{3}{2}}] \right| - \left| \frac{1}{2} [1^2 - 0^2] \right|$$

$$= \frac{2}{3} - \frac{1}{2}$$

$$= \frac{1}{6} \text{ sq. units.}$$

$$f(x) = \begin{cases} \frac{5}{2} - x & x < 2 \\ 1 & x = 2 \\ x - \frac{3}{2} & x > 2 \end{cases}$$

$$\lim_{\alpha \rightarrow 0} f(x - \alpha) = \frac{5}{2} - x$$

For $x = 2$,

$$\Rightarrow \lim_{\alpha \rightarrow 0} f(2 - \alpha) = \lim_{\alpha \rightarrow 0} \left[\frac{5}{2} - (2 - \alpha) \right]$$

$$\Rightarrow \lim_{\alpha \rightarrow 0} f(2 - \alpha) = \lim_{\alpha \rightarrow 0} \left(\frac{1}{2} + \alpha \right) = \frac{1}{2}$$

$$\lim_{\alpha \rightarrow 0} f(x + \alpha) = x - \frac{3}{2}$$

For $x = 2$,

$$\Rightarrow \lim_{\alpha \rightarrow 0} f(2 + \alpha) = \lim_{\alpha \rightarrow 0} \left[(2 + \alpha) - \frac{3}{2} \right]$$

$$\Rightarrow \lim_{\alpha \rightarrow 0} f(2 + \alpha) = \lim_{\alpha \rightarrow 0} \left(\frac{1}{2} + \alpha \right) = \frac{1}{2}$$

$$f(x) = f(2) = 1$$

$$\lim_{\alpha \rightarrow 0} f(2 - \alpha) = \lim_{\alpha \rightarrow 0} f(2 + \alpha) \neq f(2)$$

$\therefore$ The function is not continuous at $x = 2$

48.

$$S = 2 \tan^{-1} [\operatorname{cosec}(\tan^{-1} x) - \tan(\cot^{-1} x)]$$

$$\Rightarrow S = 2 \tan^{-1} [\operatorname{cosec}(\tan^{-1} x) - \tan(90 - \tan^{-1} x)]$$

$$\text{Let } \tan^{-1} x = \theta$$

$$\Rightarrow S = 2 \tan^{-1} [\operatorname{cosec} \theta - \tan(90 - \theta)]$$

$$\Rightarrow S = 2 \tan^{-1} [\operatorname{cosec} \theta - \cot \theta]$$

$$\Rightarrow S = 2 \tan^{-1} \left[\frac{1}{\sin \theta} - \frac{\cos \theta}{\sin \theta} \right]$$

$$\Rightarrow S = 2 \tan^{-1} \left[\frac{1 - \cos \theta}{\sin \theta} \right]$$

$$\Rightarrow S = 2 \tan^{-1} \left[\frac{2 \sin^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} \right]$$

$$\Rightarrow S = 2 \tan^{-1} \left[\tan \frac{\theta}{2} \right]$$

$$\Rightarrow S = \theta = \tan^{-1} x$$

49.

$$\text{Given } 3 \sin x + 4 \cos x = 5$$

$$\Rightarrow 3 \left(2 \sin \frac{x}{2} \cos \frac{x}{2} \right) + 4 \left(1 - 2 \sin^2 \frac{x}{2} \right) = 5$$

$$\Rightarrow 6 \sin \frac{x}{2} \cos \frac{x}{2} + 4 - 8 \sin^2 \frac{x}{2} = 5$$

$$\Rightarrow 6 \sin \frac{x}{2} \cos \frac{x}{2} - 8 \sin^2 \frac{x}{2} = 1$$

$$\Rightarrow 6 \sin \frac{x}{2} \cos \frac{x}{2} - 8 \sin^2 \frac{x}{2} = \sin^2 \frac{x}{2} + \cos^2 \frac{x}{2} (\because \sin^2 a + \cos^2 a = 1)$$

Dividing the equation by $\cos^2 \frac{x}{2}$

$$\Rightarrow 6 \frac{\sin \frac{x}{2} \cos \frac{x}{2}}{\cos^2 \frac{x}{2}} - 8 \frac{\sin^2 \frac{x}{2}}{\cos^2 \frac{x}{2}} = \frac{\sin^2 \frac{x}{2} + \cos^2 \frac{x}{2}}{\cos^2 \frac{x}{2}}$$

$$\Rightarrow 6 \tan \frac{x}{2} - 8 \tan^2 \frac{x}{2} = \tan^2 \frac{x}{2} + 1$$

$$\Rightarrow 6 \tan \frac{x}{2} - 9 \tan^2 \frac{x}{2} = 1$$



50. Since, $A \subset B$ and $B \subset C$, therefore $A \cup B \cup C = C$.

$$\Rightarrow n(A \cup B \cup C) = n(C).$$

$\therefore$ The cardinality (number of elements) of $A \cup B \cup C = \text{cardinality (number of elements) of } C$.

REASONING

51. Since the average of the remaining numbers is $35\frac{7}{17}$

To get an average greater than 35, the value of n must be near 69.

Let initially number of terms be 69 and the number which was erased be 'x'

$$\Rightarrow (69 \times 35 - x)/68 = 602/17$$

$$\Rightarrow 2415 - x = 2408$$

$$\Rightarrow x = 7$$

$\therefore$ The number erased was 7

52.

Friends	Time taken to cross bridge
A	1 minute
B	2 minutes
C	8 minutes
D	11 minutes

Now,

First A and B will go in 2 min, then A returns in 1 min.

Then C and D will go in 11 min. B returns in 2 min.

A & B will go in 2 min.

The total time taken would be

$$(2 + 1 + 11 + 2 + 2) = 18 \text{ min.}$$

53. Percentage of illiterate adults = $40.1\% \approx 40\%$

Percentage of literate children = $85.1\% \approx 85\%$

The ratio of the adults to that of children = $2 : 3$

Calculations:

Let the total adult be $2x$ and children be $3x$

$$\text{Total population} = 2x + 3x = 5x$$

$$\text{Total illiterate adults} = 2x \times 40/100 = 4x/5$$

$$\text{Total literate adults} = 2x - 4x/5 = 6x/5$$

$$\text{Total literate children} = 3x \times 85/100 = 51x/20$$

$$\text{Total literate} = (6x/5) + (51x/20)$$

$$\Rightarrow (24x + 51x)/20 = 75x/20$$

$$\text{Percent of the literate population} = ((75x/20) \div 5x) \times 100 = 75\%$$

54. A runs $1\frac{2}{3}$ times as fast as B

A gives B a start of 80 m

Concept used:

Time = Distance/Speed

Calculations:

Let the speed of B be x

So, speed of A = $5x/3$

Let the total distance be 'a' m

So, time taken by A to complete the total distance = $a/(5x/3) = 3a/5x$

A gives B a start of 80 m

Required distance for B = $a - 80$

Time taken by B to complete the distance = $(a - 80)/x$

A and B might reach at same time

$\therefore$ Time taken by A to complete the total distance = Time taken by B to complete the distance

$$\Rightarrow 3a/5x = (a - 80)/x$$

$$\Rightarrow 3a = 5x - 400$$

$$\Rightarrow 2a = 400$$

$$\Rightarrow a = 200$$

$\therefore$ The winning post must be 200 m far so that A and B might reach it at the same time

55. Let the age of the person's mother be 'a' units.

So, age of the person = $(2/5)a$

After 8 years, age of the person's mother = 'a + 8' units.

So age of the person after 8 years = $(a + 8)/2$

According to question:

$$(2/5)a + 8 = (a + 8)/2$$

$$\Rightarrow (2a + 40)/5 = (a + 8)/2$$

$$\Rightarrow 4a + 80 = 5a + 40$$

$$\Rightarrow a = 40$$

So, the present age of his mother is 40 years.

Hence, '40 years' is the correct answer.

56. The average speed of Mr. Kumar = 48 km/h

The time taken to cover the first 60% of the distance is 10 minutes more than the time taken to cover the remaining distance

Concept used:

Time = Distance/Speed

Calculations:

Let the total distance be x

$$60\% \text{ of the distance} = x \times 60/100 = 3x/5$$

$$\text{Time taken by Mr. Kumar to cover } 3x/5 \text{ distance} = \frac{3x}{5 \times 48} = x/80$$

$$\text{Remaining distance} = x - 3x/5 = 2x/5$$

$$\text{Time taken by Mr. Kumar to cover } 2x/5 \text{ distance} = \frac{2x}{5 \times 48} = x/120$$

According to the question,

$$\Rightarrow (x/80) - (x/120) = 10/60$$

$$\Rightarrow (3x - 2x)/240 = 10/60$$

$$\Rightarrow x = 40$$

$\therefore$ The distance of office is 40 kms

57. Taking the L.C.M. of $75/2$ & $45 = 225$ (which is total work)

Work done by A in $75/2$ minutes = 225

$$\text{Efficiency of A in 1 minute} = (225 \times 2)/75 = 6$$

Work done by B in 45 minutes = 225

$$\text{Efficiency of B in 1 minute} = 225/45 = 5$$

Let the turned off time of B be x minutes.

Efficiency of A in 1 minute $\times$ Time + Efficiency of B in 1 minute $\times$ Time = Total work

$$\Rightarrow 6 \times 30 + 5 \times x = 225$$

$$\Rightarrow 180 + 5x = 225$$

$$\Rightarrow 5x = 225 - 180$$

$$\Rightarrow x = 45/5$$

$$\Rightarrow x = 9 \text{ minutes}$$

$\therefore$ B is turned off after 9 minutes.

58. The logic is:

D	O	E	S
5	\$	3	%

S	I	T	E
%	4	#	3

Similarly,

E	D	I	T
3	5	4	#

Hence, '354#' is the correct answer.

59. Height of rain water = 5 cm

$$\text{Area of ground} = 1.5 \text{ hectares} = 1.5 \times 10^8 \text{ cm}^2$$

Concept used:

Volume = Area of base $\times$ Height

$$1 \text{ hectare} = 100000000 \text{ cm}^2 = 10^8 \text{ cm}^2$$

$$1 \text{ cm} = 1/100 \text{ m}$$



Calculation:

Volume of water that falls on the ground = $1.5 \times 10^8 \times 5$

$$\Rightarrow 75 \times 10^7 \text{ cm}^3$$

$$\Rightarrow (75 \times 10^7)/10^6 \text{ m}^3$$

$$\Rightarrow 75 \times 10 \text{ m}^3$$

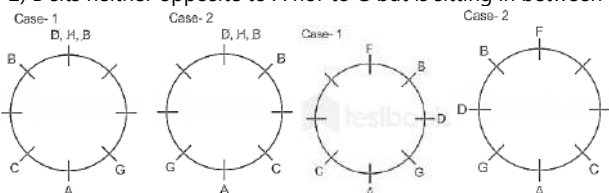
$$\Rightarrow 750 \text{ cubic meter}$$

∴ The volume of water that falls on 1.5 hectares of ground is 750 cubic meter.

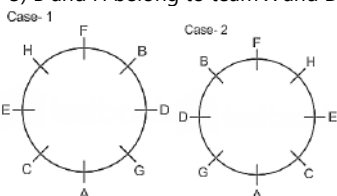
60. **Persons:** A, B, C, D, E, F, G and H

1) A sits neither opposite to D nor to H but is sitting in between C and G.

2) B sits neither opposite to A nor to G but is sitting in between F and D.



3) B and H belong to team X and D sits opposite to E



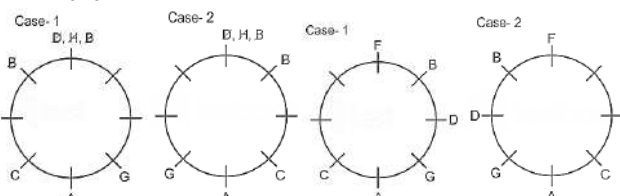
As B and H belong to team X, so is C and G.

Remaining F, E, A and D belong to team Y.

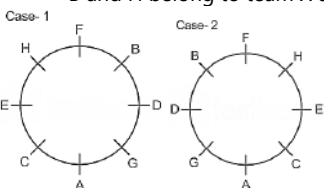
Hence, 'B, H, C and G' is the correct answer.

61. A sits neither opposite to D nor to H but is sitting in between C and G.

- B sits neither opposite to A nor to G but is sitting in between F and D



- B and H belong to team X and D sits opposite to E



As B and H belong to team X, so is C and G.

Remaining F, E, A and D belong to team Y.

So, H and C are sitting adjacent to E.

Hence, 'H and C' is the correct answer.

62. **Four students:** A, B, C and D

distributed 30 marbles among themselves

- 1) No two students got equal number of marbles.
- 2) No student got more than 10 marbles.
- 3) No student got less than 5 marbles.
- 4) A and C got odd number of marbles. It can be 7 or 9
- 5) B and D got even number of marbles. It can be 6, 8 or 10

6) A got more marbles than B, C got more marbles than D, B got more marbles than D.

$$A > B > D, C > D$$

$$A(7, 9) > B(6, 8) > D(6, 8), C(7, 9) > D(6, 8)$$

So, D got the least number of marbles.

∴ D got 6 marbles.

So, B got 8 marbles as B got more marbles than D.

A got 9 marbles as A got more marbles than B.

C got 7 marbles as C got more marbles than D.

Students	Possible number of marbles	Definite number of marbles
A	7 or 9	9
B	6 or 8	8
C	7 or 9	7
D	6 or 8	6

The number of marbles with A is 9.

Hence, '9' is the correct answer.

63. **Four students:** A, B, C and D

distributed 30 marbles among themselves

1) No two students got equal number of marbles.

2) No student got more than 10 marbles.

3) No student got less than 5 marbles.

4) A and C got odd number of marbles. It can be 7 or 9

5) B and D got even number of marbles.

It can be 6, 8 or 10

6) A got more marbles than B, C got more marbles than D, B got more marbles than D.

$$A > B > D, C > D$$

$$A(7, 9) > B(6, 8) > D(6, 8), C(7, 9) > D(6, 8)$$

So, D got the least number of marbles.

∴ D got 6 marbles.

So, B got 8 marbles as B got more marbles than D.

A got 9 marbles as A got more marbles than B.

C got 7 marbles as C got more marbles than D.

Students	Possible number of marbles	Definite number of marbles
A	7 or 9	9
B	6 or 8	8
C	7 or 9	7
D	6 or 8	6

Mean of number of marbles with B, C, D is = $(8 + 7 + 6) \div 3$

$$= 21 \div 3$$

$$= 7$$

Hence, '7' is the correct answer.

64. **ine individuals** -: Z, Y, X, W, V, U, T, S and R

Three committees-K1, K2 and K3,

1) Each candidate should serve on exactly one of the committees.

2) Among Z, Y and X none can serve on committee K1.

3) Among W, V and U none can serve on committee K2.

4) Among T, S and R none can serve on committee K3

5) It is possible that there are no members in committee K3

5) In case committee K2 is served by T and Z

Committees	Individuals who can serve	Individuals who cannot serve
K1		Z, Y, X
K2	T, Z	W, V, U
K3		T, S, R



K1 should consist of exactly one member more than committee K2.
So, K1 is served by three members as K2 is served by two members only.
So, remaining individuals will serve committee K3.
Remaining persons = $9 - (2 + 3)$
 $= 9 - 5$
 $= 4$
Hence, '4' is the correct answer.

65. **Nine individuals** :- Z, Y, X, W, V, U, T, S and R

Three committees-K1, K2 and K3,

- 1) Each candidate should serve on exactly one of the committees.
- 2) Among Z, Y and X none can serve on committee K1.
- 2) Among W, V and U none can serve on committee K2.
- 3) Among T, S and R none can serve on committee K3
- 4) It is possible that there are no members in committee K3

Committees	Individuals who can serve	Individuals who cannot serve
K1		Z, Y, X
K2		W, V, U
K3		T, S, R

Of the nine individuals, the largest number that can serve together on committee K3 = $9 - \text{Number of members who cannot serve K3}$
 $= 9 - 3$
 $= 6$
Hence, '6' is the correct answer.

66. The pattern followed here is:

Alphabets	A	B	C	D	E	F	G	H	I	J	K	L	M
Positional value	1	2	3	4	5	6	7	8	9	10	11	12	13
Positional value	26	25	24	23	22	21	20	19	18	17	16	15	14
Alphabets	Z	Y	X	W	V	U	T	S	R	Q	P	O	N

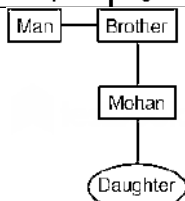
According to the alphabetical positions of the letters,

E $\xrightarrow{+2}$ G $\xrightarrow{+2}$ I $\xrightarrow{+2}$ K $\xrightarrow{+2}$ M
L $\xrightarrow{+2}$ L $\xrightarrow{+2}$ L $\xrightarrow{+2}$ L $\xrightarrow{+2}$ L
F $\xrightarrow{+2}$ H $\xrightarrow{+2}$ J $\xrightarrow{+2}$ L $\xrightarrow{+2}$ N
A $\xrightarrow{+2}$ A $\xrightarrow{+2}$ A $\xrightarrow{+2}$ A $\xrightarrow{+2}$ A

Hence, 'KLLA' is the correct answer.

67. According to given conditions,

Symbol in Diagram	Meaning
○	Female
□	Male
==	Married couple
—	Siblings
	Difference of a generation



So, the gentleman is Mohan's Uncle.
Hence, 'Uncle' is the correct answer.

68. It was 9.35 AM in Garvita's watch, which kept correct time.
Manya informed her that the last bus left the bus stop at 9.25 am.
Manya's watch is 5 min fast.
So, the actual time when the bus left the bus stop = $9:25 - 0:05 \text{ AM}$
 $= 9:20 \text{ AM}$
The frequency of the bus is every 20 min.
So, the next bus will arrive at bus stop at 9:40 AM.
As it was 9.35 AM in Garvita's watch, she should wait 5 more minutes to catch the next bus.
Hence, '5 min' is the correct answer.
Was the solution helpful? Yes/No

69. Total number of notes of Rs. 20 and Rs. 50 = 324
Total Sum = Rs. 12450

Concept used:

Value of note $\times$ Number of total notes = Total amount

Calculation:

Let the number of Rs. 20 notes be x.

No. of Rs. 50 notes = $324 - x$

$\Rightarrow 20 \times x + 50 \times (324 - x) = 12450$

$\Rightarrow 20x + 16200 - 50x = 12450$

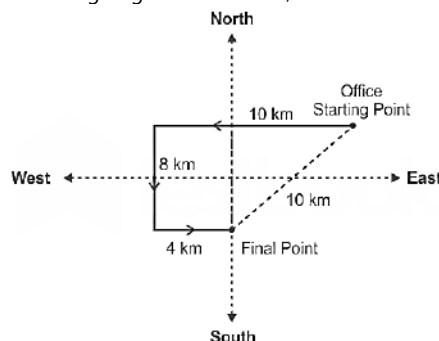
$\Rightarrow -30x = 12450 - 16200$

$\Rightarrow 30x = 3750$

$\Rightarrow x = 125$

$\therefore$ The number of Rs. 20 notes is 125.

70. According to given conditions,



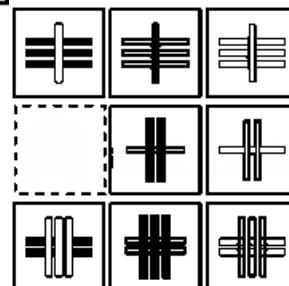
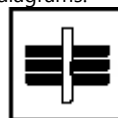
Distance between his starting point and final point = $\sqrt{8^2 + 6^2} \text{ km}$
 $= \sqrt{64 + 36} \text{ km}$
 $= \sqrt{100} \text{ km}$
 $= 10 \text{ km}$

Hence, '10 km' is the correct answer.

71. Let us divide the given set of images in a systematic manner.
Along the rows and columns we can see a particular pattern is maintained.

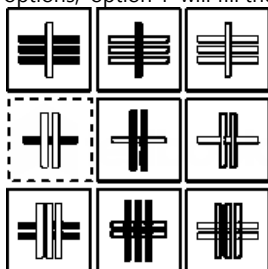
It makes our task easy to figure out which of the given options will fill the missing cell.

- 1) Three color code is maintained, white, blue, yellow
- 2) 3 pattern are maintained thrice throughout the diagram with
- 3) Each pattern has yellow, blue and white in the behind one lines in diff diagrams.





So, from the given options, 'option 1' will fill the missing cell.



Hence, 'option 1' is the correct answer.

72.

P is				
\$	#	@	%	*
≥	<	>	≤	=
To Q				

$K \# L \rightarrow K < L$

$L \% M \rightarrow L \leq M$

$M * N \rightarrow M = N$

$N \# O \rightarrow N < O$

Combining all the statements together, we get:

$K < L \leq M = N < O$

I. $K \# M \rightarrow K < M \rightarrow$ True (As, $K < L \leq M \rightarrow K < M$)

II. $K * M \rightarrow K = M \rightarrow$ False (As, $K < L \leq M \rightarrow K < M$)

III. $L \% O \rightarrow L \leq O \rightarrow$ False (As, $L \leq M = N < O \rightarrow L < O$)

Hence, 'I only' is the correct answer.

73. In all except, 9 Q A the second element is **second to the left** of first element and third element is **second to the right** of first element.

W 1 R % 4 J E # 7 M T 2 I 9 B H 3 A \$ 9 F Q 5 D G 6 U S P

All options except '9 Q A' follows the same pattern of arrangement as highlighted in the above series.

Hence, '9 Q A' is the odd one out.

74. If there are no dancers that aren't slim $\rightarrow$ Implies all dancers are slim.

No singers that aren't dancers $\rightarrow$ All singers are dancers.

Thus from the above two it can be concluded that all singers are dancers and all dancers are slim, so all singers are slim.

1) There is not one slim person that isn't a dancer. $\rightarrow$ False (It is possible but not definite)

2) All singers are slim $\rightarrow$ True (It is definite)

3) Anybody slim is also a singer $\rightarrow$ False (It is possible but not definite)

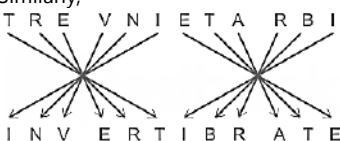
4) None of the above. $\rightarrow$ False (As option 2 is true)

Hence, 'All singers are slim' is the correct answer.

74. The logic is:



Similarly,



Hence, 'INVERTIBRATE' is the correct answer.

75. Sum of ages of Anu and Bhanu is 10 years more than sum of ages of Bhanu, Chanu and Dhanu.

$Bhanu + Chanu + Dhanu + 10 = Anu + Bhanu$

$\Rightarrow Chanu + Dhanu + 10 = Anu$ < i >

Average age of Chanu and Dhanu is 19 years.

$(Chanu + Dhanu) / 2 = 19$

$Chanu + Dhanu = 19 \times 2$ years

$Chanu + Dhanu = 38$ years < ii >

Putting the value of < ii > in < i >, we get:

$38 + 10 = Anu$

So, age of Anu is 48 years

Dhanu is 10 years elder than Chanu.

$Dhanu - 10 = Chanu$ < iii >

Putting the value of < iii > in < ii >, we get:

$Dhanu - 10 + Dhanu = 38$

$\Rightarrow 2Dhanu = 38 + 10$

$\Rightarrow 2Dhanu = 48$

$\Rightarrow Dhanu = 24$ years

Average age of Anu and Dhanu = $(48 + 24) \div 2$ years

= $72 \div 2$ years

= 36 years

Hence, '36 years' is the correct answer.

76. Let total number of candidates from each state be 100 unit.

Selected candidate in Maharashtra from the total appeared candidates = $(9/100) \times 100 = 9$ unit.

Selected candidate in Tripura from the total appeared candidates = $(12/100) \times 100 = 12$ unit.

In Tripura number of selected candidates more than that of Maharashtra = 102

$\Rightarrow (12 - 9)$ unit = 102

$\Rightarrow 3$ unit = 102

$\Rightarrow 1$ unit = $102/3$

$\Rightarrow 1$ unit = 34

$\Rightarrow (1 \times 100)$ unit = 34×100

$\Rightarrow 100$ unit = 3400

$\therefore$ The number of candidates appeared from each state was 3400.

77. Let the sum of amount be 50 unit. (where 50 = product of denominator of given fraction 5×10)

Salary given to his mother = $(2/5) \times 50 = 20$ unit.

Remaining salary after gave to his mother = $50 - 20 = 30$ unit.

Electronics gadgets and monthly expenses from the remaining = $(3/10) \times 30 = 9$ unit.

Remaining amount which transferred to his friend's account = $30 - 9 = 21$ unit

Sum of amount he gave to his mother and his friend's account = 41000

$\Rightarrow (20 + 21)$ unit = 41000

$\Rightarrow 41$ unit = 41000

$\Rightarrow 1$ unit = $41000/41$

$\Rightarrow 1$ unit = 1000

$\Rightarrow (1 \times 50)$ unit = 1000×50

$\Rightarrow 50$ unit = 50000

$\therefore$ The monthly salary of Shiva is Rs. 50000 .

78. i. $A * B$ means $\rightarrow$ A and B are of the same age

ii. $A - B$ means $\rightarrow$ B is younger than A $\rightarrow$ A is older than B

iii. $A + B$ means $\rightarrow$ A is younger than B

A is		
*	+	-
=	<	>
To B		

Sachin * Madan - Reena means?

Sachin = Madan > Reena means Reena is the youngest.

Hence, 'Reena is youngest' is the correct answer.



79. i. $A * B$ means $\rightarrow$ A and B are of the same age
 ii. $A - B$ means $\rightarrow$ B is younger than A $\rightarrow$ A is older than B
 iii. $A + B$ means $\rightarrow$ A is younger than B

A is		
*	+	-
=	<	>
To B		

80. $X + Y + Z \rightarrow X < Y < Z$
 1) $Y - X - Z \rightarrow Y > X > Z \rightarrow Z < X < Y$
 2) $Z - Y - X \rightarrow Z > Y > X \rightarrow X < Y < Z$
 3) $Z - X - Y \rightarrow Z > X > Y \rightarrow Y < X < Z$
 4) $X - Y - Z \rightarrow X > Y > Z \rightarrow X > Y > Z$
 So, $X + Y + Z$ is same as $Z - Y - X$.
 Hence, ' $Z - Y - X$ ' is the correct answer.

81. The logic is:
 56 58 62 70 86 118 182
 $+2$ $+4$ $+8$ $+16$ $+32$ $+64$
 $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 $+2^1$ $+2^2$ $+2^3$ $+2^4$ $+2^5$ $+2^6$
 Hence, '84' is the correct answer.

82. The logic is:
 $2^2 = 4$ $3^2 = 9$
 123
 $1^2 = 1$

Similarly,
 $3^2 = 9$ $2^2 = 4 = ?$
 232
 $2^2 = 4$

Hence, '4' is the correct answer.

83. Students who failed = $(100 - 78)\% = 22\%$
 22% of total students = 176
 1% of total students = $176/22 = 8$
 34% of total students = $8 \times 34 = 272$
 $\therefore$ The number of students who got first class are 272.

84. The logic is:
 Column 1: $2 \times 1 + 1 = 3$
 Column 2: $14 \times 7 + 7 = 105$
 Similarly,
 Column 3: $? \times 9 + 9 = 117$
 $\Rightarrow ? \times 9 = 117 - 9$
 $\Rightarrow ? \times 9 = 108$
 $\Rightarrow ? = 108 \div 9$
 $\Rightarrow ? = 12$
 Hence, '12' is the correct answer.

85. The logic is:
 $96 \div 2 = 48$; $48 \div 2 = 24$
 Similarly,
 $64 \div 2 = 32$; $32 \div 2 = 16$
 Hence, '16' is the correct answer.

86. As no two adjacent faces can have the same colour, the pair of two opposite faces can have the same colour.
 $\therefore$ Minimum numbers of colours required = $6 \div 2 = 3$
 Hence, '3' is the correct answer.

87. Let the correct time of train arrival be t and distance to reach the station be d .

$$\Rightarrow d = 4 \times (t + 6)/60 \quad \text{---(1)}$$

$$\Rightarrow d = 5 \times (t - 6)/60 \quad \text{---(2)}$$

From equations (1) and (2),

$$\Rightarrow 4 \times (t + 6)/60 = 5 \times (t - 6)/60$$

$$\Rightarrow 4t + 24 = 5t - 30$$

$$\Rightarrow 5t - 4t = 30 + 24$$

$$\Rightarrow t = 54 \text{ min}$$

Putting the value of t in equation (1),

$$\Rightarrow d = 4 \times (54 + 6)/60$$

$$\Rightarrow 4 \times 60/60$$

$$\Rightarrow 4 \text{ km}$$

$\therefore$ The distance covered by him to reach the station is 4 km.

88. Increase in numerator = 25%

Decrease in denominator = 20%

New value = $5/4$

Concept used:

$$x\% \text{ increase} = (100 + x)/100$$

$$x\% \text{ decrease} = (100 - x)/100$$

Calculation:

Let the original fraction be x/y .

$$\text{Change in numerator} = (100 + 25)/100 = 5/4$$

$$\text{Change in denominator} = (100 - 20)/100 = 4/5$$

$$\Rightarrow (x \times 5/4)/(y \times 4/5) = 5/4$$

$$\Rightarrow (5x \times 5)/(4y \times 4) = 5/4$$

$$\Rightarrow 25x/16y = 5/4$$

$$\Rightarrow x/y = 4/5$$

$\therefore$ The original value of the fraction is $4/5$.

89. **Five friends:** Amar, Kapil, Sarvesh, Rohan and Nagesh

Different colours: Red, Yellow, Blue, White, and Green, while they were going to attend a party. These colours are not in the order.

Different hobbies: reading, playing, outing, singing and writing.

1) Kapil, who likes singing, does not wear Yellow shirt.

2) Sarvesh wears Red shirt and he does not like reading or writing.

3) Nagesh likes playing and he does not wear Blue or Yellow shirt.

4) Amar likes writing and Rohan does not wear Yellow or Green shirt.

Friends	Colour	Hobbies
Amar		Writing
Kapil	Yellow	Singing
Sarvesh	Red	Reading, Writing
Rohan	Yellow, Green	
Nagesh	Blue, Yellow	Playing

Amar wears Yellow shirt and Rohan likes Reading.

Sarvesh likes outing.

Friends	Colour	Hobbies
Amar	Yellow	Writing
Kapil	Yellow	Singing
Sarvesh	Red	Outing
Rohan	Yellow, Green	Reading
Nagesh	Blue, Yellow	Playing

The colour of Kapil's shirt may be Green or Blue or White.



The colour of Rohan's shirt may be White or Blue.
The colour of Nagesh's shirt may be White or Green.

Friends	Colour	Hobbies
Amar	Yellow	Writing
Kapil	Green/Blue/ White	Singing
Sarvesh	Red	Outing
Rohan	Blue/ White	Reading
Nagesh	Green/White	Playing

The colour of Kapil's shirt cannot be answered as data is insufficient.
Hence, 'Insufficient data to answer' is the correct answer.

90. **Five friends:** Amar, Kapil, Sarvesh, Rohan and Nagesh
Different colours: Red, Yellow, Blue, White, and Green, while they were going to attend a party. These colours are not in the order.
Different hobbies: reading, playing, outing, singing and writing.
1) Kapil, who likes singing, does not wear Yellow shirt.
2) Sarvesh wears Red shirt and he does not like reading or writing.
3) Nagesh likes playing and he does not wear Blue or Yellow shirt.
4) Amar likes writing and Rohan does not wear Yellow or Green shirt.

Friends	Colour	Hobbies
Amar		Writing
Kapil	Yellow	Singing
Sarvesh	Red	Reading, Writing
Rohan	Yellow, Green	
Nagesh	Blue, Yellow	Playing

Amar wears Yellow shirt and Rohan likes Reading.
Sarvesh likes outing.

Friends	Colour	Hobbies
Amar	Yellow	Writing
Kapil	Yellow	Singing
Sarvesh	Red	Outing
Rohan	Yellow, Green	Reading
Nagesh	Blue, Yellow	Playing

The colour of Kapil's shirt may be Green or Blue or White.
The colour of Rohan's shirt may be White or Blue.
The colour of Nagesh's shirt may be White or Green.

Friends	Colour	Hobbies
Amar	Yellow	Writing
Kapil	Green/Blue/ White	Singing
Sarvesh	Red	Outing
Rohan	Blue/ White	Reading
Nagesh	Green/White	Playing

So, Amar likes Writing.

Hence, 'Amar' is the correct answer.

COMPUTER

91. He correct answer is $x' + y$.

To find the equivalent of $x'y' + xy + x'y$,

First take x' common from $x'y' + x'y$

$$= xy + x'(y + y')$$

$$\text{Use } A + A' = 1$$

$$= xy + x' \cdot 1$$

$$\text{Use } (A+BC) = (A+B)(A+C)$$

$$= (x + x')(y + x')$$

$$\text{Use } A + A' = 1$$

$$= 1 \cdot (x' + y)$$

- Hence $(x' + y)$ is the answer.

92. The correct answer is **Primary Memory**

- The memory unit which directly communicates with the CPU is known as **Primary Memory**.

93. The correct answer is **More, Slower**.

94. The correct answer is **(11101010.001)₂**

95. The correct answer is **None of the above**

96. The correct answer is **All of the above**.

One Exabyte is equal to:

$$= 10^{18} \text{ Bytes.}$$

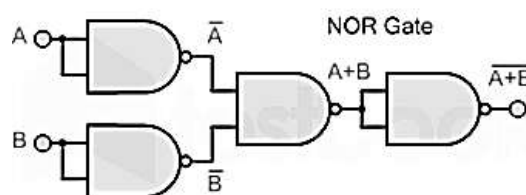
$$= 1000 \times 1 \text{ PB}$$

$$= 1 \text{ ZB} / 1000$$

97. The correct answer is **4**

EXPLANATION:

- As per the question, we need to create a **NOR gate** (Question Diagram) using **NAND gates**.
- The boolean expression for NAND is $Y = A \cdot B \Rightarrow Y = A \cdot B$
- The boolean expression for NOR is $Y = A + B \Rightarrow Y = A + B$
- So, using the above boolean expression, the following is the possible diagram:



98. The correct answer is **CPU cycle**.

- The time required for the fetching and execution of one simple machine instruction is **CPU cycle time**.
- The fundamental sequence of steps that a CPU performs is called CPU Cycle, it is also known as the "**fetch-execute cycle**".
- In simpler CPUs, the instruction cycle is executed sequentially, each instruction being processed before the next one is started.
- In most modern CPUs, the instruction cycles are instead executed concurrently, and often in parallel, through an instruction pipeline
- A **clock cycle**, or simply a "cycle," is a single electronic pulse of a CPU.
- During each cycle, a CPU can perform a basic operation such as **fetching an instruction, accessing memory, or writing data**.



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99. The correct answer is $x + y$.

To solve $X + X'Y$

Use $X.1 = X$,

$= X.1 + X'Y$

Use $1 + Y = 1$

$= X(1 + Y) + X'Y$

$= X + XY + X'Y$

$= X + Y(X + X')$

Use $X + X' = 1$

$= X + Y.1$

Use $Y.1 = Y$

$= X + Y$

- Hence answer is $X + Y$.

100. The correct answer is FFFF

101. The correct answer is "Super!"

102. The correct answer is "Carelessness".

103. The correct answer is "Truant".

104. The correct answer is "Punctuation error".

105. The correct answer is "Lessen".

106. The correct answer is "Sensible".

107. The correct answer is 'Difficult to understand'.

108. The correct answer is "To confirm".

109. The correct answer is Option 2: "To be patient".

110. The correct answer is "Something good but not recognised at first".

111. The correct answer is "of".

112. The correct answer is "to".

113. The correct answer is "sat".

114. The correct answer is "Covering letter".

115. The correct answer is "Family history".

116. The correct answer is "I prefer coffee to tea".

117. The correct answer is "Lichen is an independent plant".

118. The correct answer is "that sharing of things help easy growth".

119. The correct answer is "Peeling".

120. The correct answer is "Compress".





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